Two Spectral–Spatial Implicit Neural Representations for Arbitrary-Resolution Hyperspectral Pansharpening

Lin He, *Member, IEEE*, Zhou Fang, Jun Li, *Fellow, IEEE*, Jocelyn Chanussot, *Fellow, IEEE*, and Antonio Plaza, *Fellow, IEEE*

***Abstract*— Standard hyperspectral (HS) pansharpening utilizes panchromatic (PAN) images to improve the connected low (spatial) resolution HS (LRHS) images to the spatial resolutions of PANs, while arbitrary-resolution HS (ARHS) pansharpening aims to use PANs to enhance LRHS images to any desired spatial resolutions. For the challenging task of ARHS pansharpening, one of the major obstacles is how to generalize the single pansharpening model learned under predetermined training scales to any pansharpening scales for future data. As implicit neural representations (INRs) have the potential to approximate continuous functions, they offer a possible alternative way to naturally resolve ARHS pansharpening. In this article, we develop two spectral–spatial INRs for ARHS pansharpening: one is a naive spectra–spatial pansharpening INR (NaivePINR) and the other is a dynamic spectra–spatial pansharpening INR (DynamicPINR). The former builds a novel spectral–spatial encoding to produce spectral–spatial priors of observed scenes and uses a spectral–spatial query mapping to reconstruct fine spectral details and spatial details. The latter establishes an innovative twofold tuning mechanism to dynamically adjust both the spectral–spatial encoding and the spectral–spatial query mapping. The experimental results on serval datasets verify the excellent performances of the proposed pansharpening INRs.**

***Index Terms*— Arbitrary-resolution pansharpening, dynamic implicit neural representation (INR), encoding, hyperspectral (HS) images, query mapping, spectral–spatial INRs.**

I. Introduction

IMAGING spectrometers are able to acquire remote sensing images with many continuous spectral bands. Those hyperspectral (HS) images observe the spectral behavior of the ground [1], [2], [3] and thus offer great potentials for improving many remote sensing applications [3], [4], [5], [6]. However, in HS data, high spectral resolutions tend to inherently restrain their spatial resolutions, which prevents such data to highly resolve spatial details. To overcome the obstacle, a feasible way is to rely on HS pansharpening, which seeks to fuse low (spatial) resolution HS (LRHS) images with the registered high (spatial) resolution panchromatic (PAN) images to enhance LRHS images to the spatial resolutions of their PAN counterparts. The resulting data are high (spatial) resolution HS (HRHS) images with high spatial resolutions and high spectral resolutions simultaneously [2]. Hereinafter, such a pansharpening, which uses PANs to promote LRHS images to the spatial resolutions of PANs, is called *standard HS pansharpening*.

In fact, the HS pansharpening technique evolves from classical multispectral (MS) pansharpening where PAN images are utilized to enhance the spatial resolution of the associated MS images [7]. Some traditional MS pansharpening methods can be lent to HS pansharpening, such as component substitution (CS), multiresolution analysis (MRA), and variational optimization (VO) [8], [9], [10], [11], [12], [13], [14], [15], [16]. Relative to MS pansharpening, HS pansharpening task is faced with some special problems, such as great spectral resolution difference between the HS data and the PAN data and recovering spatial details in many continuous bands by collaborating with merely one PAN band. Bayesian approaches aim to establish a statistical framework that models the probability distributions of HRHS images conditioned by observed LRHS images and PAN images and further associates certain priors with HRHS images to resolve pansharpened data [17], [18]. Matrix factorization approaches suppose the subspaces of HRHS images and then use dictionary learning and knowledge such as sensor characteristics to infer the subspaces from the observed HS and PAN data to yield pansharpening results [19].

Convolutional neural network (CNN)-based pansharpening is currently witnessing a great surge of interesting in the remote sensing community [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], which possesses the potential to refine patterns and structures from the complex data without the requirement of explicit model assumptions. The technique originates from several pioneer

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Lin He and Zhou Fang are with the School of Automation Science and Engineering and Guangdong Engineering Technology Research Center of Unmanned Aerial Vehicle System, South China University of Technology, Guangzhou 510640, China (e-mail: helin@scut.edu.cn).

Jun Li is with the School of Computer Science and Hubei Key Laboratory of Intelligent Geo-Information Processing, China University of Geosciences, Wuhan 430078, China (e-mail: lijuncug@cug.edu.cn).

Jocelyn Chanussot is with Inria, CNRS, Grenoble INP, LJK, University of Grenoble Alpes, 38000 Grenoble, France (e-mail: jocelyn.chanussot@grenoble-inp.fr).

Antonio Plaza is with the Hyperspectral Computing Laboratory, University of Extremadura, 10003 Cáaceres, Spain (e-mail: aplaza@unex.es).

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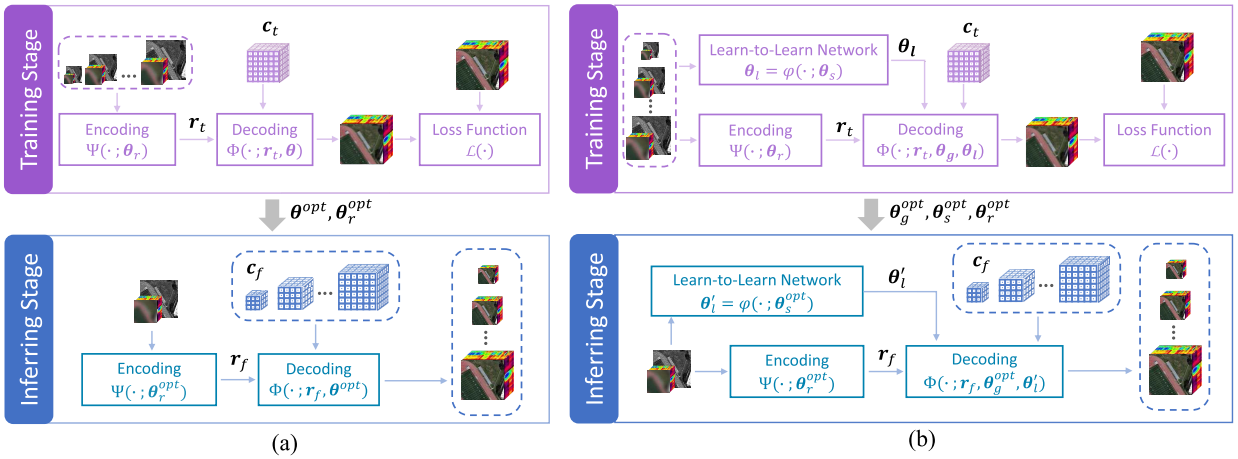


Fig. 1. Pipelines of spectral–spatial pansharpening INRs. (a) NaivePINR. (b) DynamicPINR.

works. Masi et al. [20] and Scarpa et al. [21] concatenated the pre-interpolated MS and PAN to directly feed a lightweight CNN and further develop a target-adaptive fine-tune strategy to enhance the network, respectively. Wei et al. [22] and Yuan et al. [23] introduced multiscale kernels to residual network for pansharpening. Rao et al. [24] and He et al. [25] developed detail injection structured pansharpening CNNs. More recently, some CNN-based methods have been exploited for HS pansharpening. Li et al. [26] designed a Laplacian pyramid network to coarsely spatially enhance each band and then tune details by a guided filter and a gain matrix. Xie et al. [27] proposed a high-frequency network estimating the relevant constraint term of Sylvester equation to enhance the quality of reconstructed results. He et al. [28], [29] developed spectral predictive and spectral-fidelity oriented CNN structures to specifically alleviate notorious spectral distortions. Qu et al. [30] designed a cross-guided pyramid attention CNN to synthesize the spectral attention and the spatial attention. Guarino et al. [31], [32] fused principal components of LRHS data with PAN data under a full-resolution loss [33] and then further proposed a sequential band-wise adaptive scheme to propagate a CNN model along the spectral dimension, respectively. It is noteworthy that some CNNs were developed for HS pansharpening-related tasks such as HSI and MSI fusion. For example, Zhang et al. [34] devised a three-stage CNN to sequentially perform spatial reconstruction and spectral reconstruction for HSI and MSI fusion. Sun et al. [35] developed a three-branch network consisting of dual pyramid subnetworks and a spectralwise transformer subnetwork to integrate spatial and spectral information of HSI and MSI data.

Arbitrary-resolution HS (ARHS) pansharpening is an emerging field of HS pansharpening. In contrast to standard pansharpening task, *ARHS pansharpening* utilizes PAN images to pansharpen HS images to any desired spatial resolutions [36], [37], [38], [39], [40]. Such a new HS pansharpening technique is flexible in adjusting the spatial resolutions of the resulting HRHS images and therefore provides great potentials in many applications [36], such as enhancing thematic mapping [36], [41], [42], [43], [44], object detection [36], [45], [46], and land use/cover classification [36], [43], [47]. However, ARHS pansharpening is more challenging than standard HS pansharpening. In practical applications, pansharpening neural networks are usually learning-based, which means that a pansharpening process involves a training stage and a subsequent inferring stage. In the training stage, a pansharpening model is learned using reduced-resolution data; then, in the inferring stage, incoming data are sharpened with the trained model. For standard HS pansharpening, the pansharpening scale (i.e., the ratio of the spatial resolution of target HRHS data and that of the LRHS data) is fixed across the training and the inferring stages under the stipulation of scale invariance of the trained model, which ensures the generalization of the trained pansharpening model to the inferring stage. However, ARHS pansharpening has to overcome some unprecedented dilemmas. For ARHS pansharpening task, there may be many possible desired pansharpening scales in the inferring stage, which means that the CNN model trained under given scales in the training stage has to be generalized to deal with unexpected upcoming scale requirements in the inferring stage. Therefore, a major dilemma ARHS pansharpening that has to overcome is how to generalize a single pansharpening model learned under certain training scales to reconstruct target HRHS images for future data at any resolutions, which is beyond the training scales [36]. He et al. [36] used a location-sharing mechanism to establish a novel two-stage relay-based ARHS pansharpening CNN (ARHS-CNN). Furthermore, they built a novel spatial mask-sharing mechanism into a variable subpixel convolution-based CNN (VSPC-CNN) to accomplish the ARHS pansharpening task [37]. Over recent years, implicit neural representations (INRs) draw some attention in the fields of computer vision and image processing, owing to their potentials to fulfill fine reconstruction [48], [49], [50], [51], [52], [53], [54]. INRs employ neural networks to parametrically approximate continuous functions, which means a potential ability to yield estimations on continuous coordinates. Therefore, they offer possible alternative clues for ARHS pansharpening to reconstruct HRHS images at any

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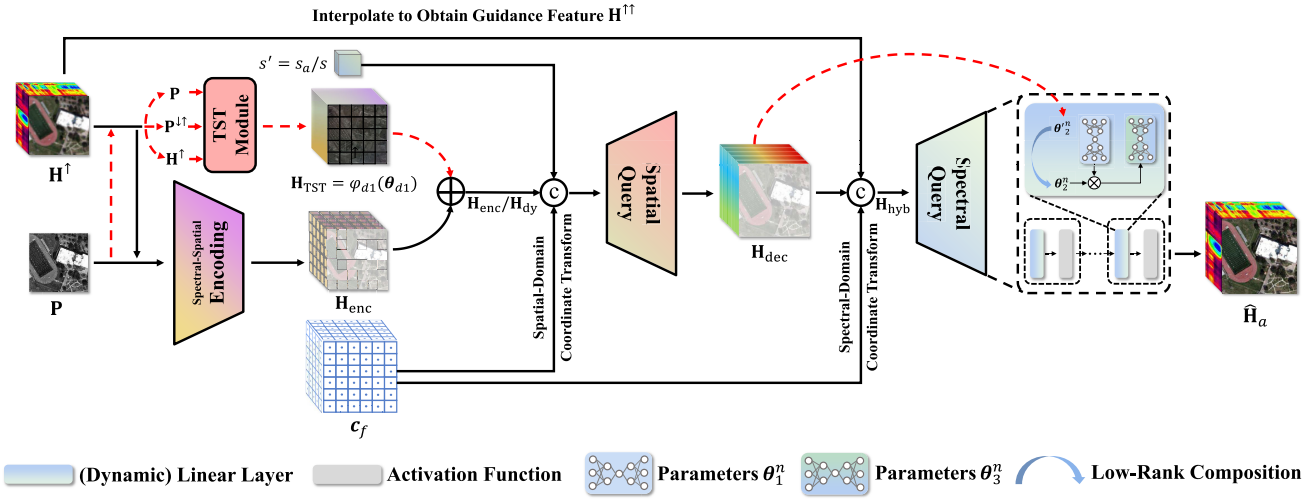


Fig. 2. Overall architectures of our spectral–spatial pansharpening INRs for ARHS pansharpening, where red dashed lines and connections are used for our DynamicPINR. ⊕ denotes the element-wise sum, ⊗ denotes the element-wise product, and © denotes the concatenation operation.

spatial resolution using a single pansharpening model trained under fixed training scales.

In this article, we develop two novel spectral–spatial pansharpening INRs for ARHS pansharpening, including a naive spectral–spatial pansharpening INR (NaivePINR) and a dynamic spectral–spatial pansharpening INR (DynamicPINR). Their components, including spectral–spatial encoding, spectral–spatial query mapping, and dynamic adjustments, are newly designed. Our pansharpening INRs aim at fulfilling ARHS pansharpening and therefore have advantages over standard HS pansharpening in flexibly adjusting the spatial resolution of target HRHS data and providing possibilities for broader applications in remote sensing tasks, as aforementioned. In addition, our pansharpening INRs are obviously different from those HSI and MSI fusion methods such as [34] and [35], which render direct resolution enhancement structures. Our INRs use latent space priors and parameterized INR networks to approximate continuous HS scene functions in encoding–decoding contexts, so as to resolve ARHS pansharpening. More specifically, innovative contributions of our work can be identified as follows.

1) *A Naive Spectral–Spatial Pansharpening INR:* We first build a formulated framework of an encoding-query process (as seen in Section IV-A). Then, following the overall design, NaivePINR is developed. The novel INR is trunk only structured, mainly comprising a subnetwork of spectral–spatial encoding to refine spectral–spatial priors of observed scenes; and meanwhile a subnetwork of spectral–spatial query mapping to build coordinate-data connections for not only spatial coordinates but also spectral coordinates, thus reconstructing fine spatial details and fine spectral details to fulfill the ARHS reconstruction.
2) *A Dynamic Spectral–Spatial Pansharpening INR:* To further enhance the naive spectral–spatial INR, we establish a formulated framework for a dynamic encoding query (as seen in Section IV-A). Then, based on such an overall design, an innovative twofold dynamic pansharpening INR is developed. In the process of spectral–spatial encoding, global spatial dependencies of the input scene are used to dynamically adjust the encoding. In the process of spectral–spatial query mapping, a spatial query is used to dynamically tune the parameters of multilayer perceptrons (MLPs) of spectral query.
3) *A Spectral–Spatial Encoding:* We design a novel encoding to effectively learn the spectral–spatial latent space in our pansharpening INRs. During the encoding, preliminary spectral–spatial fusion, spectral–spatial dual-attention, and spectral–spatial dual-adaptiveness are cooperated to not only remove the redundancy of input data but also to obtain effective spectral–spatial latent codes connected to specific HS scenes.
4) *A Spectral–Spatial Query Mapping:* We design this novel mapping to decode latent codes which contain spectral–spatial priors of input PAN and LRHS data, so as to have a representation of HRHS images in a spectral–spatial continuous manner. In this mapping, not only spectral coordinates but also spatial coordinates are directly used to query data, which offers a potential to recover fine spectrum characteristics to improve spectral preserving and, meanwhile, synthesize fine spatial details in high quality.
5) *A Twofold Dynamic Tuning Mechanism:* We develop an innovative twofold dynamic tuning mechanism to enhance the spectral–spatial INR's ability to refine inherent information in data. The dynamic tuning includes operations on both the spectral–spatial encoding and the spectral–spatial query. In the dynamic spectral–spatial encoding, we newly design a texture similarity transfer (TST) module to strengthen global texture similarities; In the dynamic spectral–spatial query mapping, we present an innovative design that uses spatial query to guide a low-rank-based spectral query in order to enhance the reconstruction of spectral–spatial details.

The remainder of this article is organized as follows. Section II presents a general description of background of INRs. The overall designs of spectral–spatial pansharpening INRs are given in Section III. The details of developing our naive

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spectral–spatial pansharpening INR are given in Section IV. Section V presents the design of our dynamic spectral–spatial pansharpening INR in detail. Relevant experiments on serval datasets and their results are reported in Section VI. Section VII concludes our work in this article.

II. Background of INRs

INRs are built on an idea that neural networks can parametrically estimate complex continuous-time (CT) signals with the observed data. Suppose a 1-D CT signal s as follows:

$$s = \zeta(c) \tag{1}$$

where $c \in \mathbb{R}$ denotes a 1-D coordinate and $\zeta(\cdot)$ denotes the continuous function associated with the CT signal. Then, we could use a neural network to parameterize the signal

$$s = \phi(c; \boldsymbol{\theta}) \tag{2}$$

where $\phi(\cdot\,; \boldsymbol{\theta})$ denotes the neural network and $\boldsymbol{\theta}$ comprises the parameters of the network.

INRs are originally designed for reconstruction of 3-D objects [48], [49], [50], [51], [52], [53]. In that application, theoretically, each specific object is related to a specific function representation and thus to a specific INR, which means the requirement of learning an independent INR for each of all the possible objects, that is, a learning result is unapplicable to upcoming unseen objects. Therefore, a function space needs to be represented with a neural network for inferring or generalization. The function space is a latent space where each object corresponds to a latent code, i.e., the prior connected to the object. Accordingly, a neural network can represent the latent space by involving the prior, which can be formulated as

$$s = \phi(c; \boldsymbol{r}, \boldsymbol{\theta}) \tag{3}$$

where $\phi(\cdot\,; \cdot, \boldsymbol{\theta})$ stands for the function space. Once the prior $\boldsymbol{r}$ is determined, $\phi(\cdot\,; \boldsymbol{r}, \boldsymbol{\theta})$ is associated with a specific signal. An effective way to build such a kind of INRs is to employ a encoder–decoder structure, where the encoder is used to learn the latent space and then accordingly regress the latent code connected to a specific object, while the decoder aims to reconstruct the input from the latent code in a continuous manner.

In order to enable the decoding to recreate the input based on the latent code, the coordinates of the input and the latent code are usually integrated using an MLP. MLPs stand for a class of forward neural networks, which usually comprise fully connected layers and flatten input vectors, and are often used for decoding in INRs since they are able to directly incorporate the data in different attributes.

III. Overall Designs of Our Pansharpening INRs

HS images contain not only spatial information but also spectral information, which implies that INRs for HS pansharpening should accommodate both spatial and spectral considerations.

For the HS pansharpening task, an INR is usually learning-based and therefore is composed of two consecutive stages: one is the training stage, while the other is the inferring stage. The training stage of the pansharpening INR can be formulated as

$$\boldsymbol{\theta}^{\text{opt}}, \boldsymbol{\theta}_r^{\text{opt}} = \arg\min_{\boldsymbol{\theta}, \boldsymbol{\theta}_r} \mathcal{L}(\Phi(\boldsymbol{c}_t; \boldsymbol{r}_t = \Psi(\widetilde{\mathbf{H}}_t^{\uparrow}, \widetilde{\mathbf{P}}_t; \boldsymbol{\theta}_r), \boldsymbol{\theta}), \mathbf{H}_t) \tag{4}$$

where $\boldsymbol{c}_t = (x, y, b)$ denotes the samplings of continuous spectral–spatial coordinates of the training HS data, $\boldsymbol{r}_t$ represents the priors connected to the training data, $\Psi(\cdot)$ refers to the parameterized subnetwork to yield the priors (i.e., the spectral–spatial latent codes), $\boldsymbol{\theta}_r$ is the parameter of the encoding, $\Phi(\cdot)$ refers to the subnetwork of query mapping for decoding, and $\boldsymbol{\theta}$ is the parameter of the network $\Phi(\cdot)$. Once the parameters $\boldsymbol{\theta}$ are trained with the discretely represented samples, the neural network $\Phi(\cdot\,; \cdot, \boldsymbol{\theta})$, which approximates a continuous function ζ, is determined, which can be formulated as

$$\widehat{\mathbf{H}}_a = \Phi(\boldsymbol{c}_f; \boldsymbol{r}_f = \Psi(\mathbf{H}^{\uparrow}, \mathbf{P}; \boldsymbol{\theta}_r^{\text{opt}}), \boldsymbol{\theta}^{\text{opt}}) \tag{5}$$

where $\boldsymbol{c}_f = (x, y, b)$ denotes the samplings of continuous spectral–spatial coordinates of the inferring HS data and $\boldsymbol{r}_f$ represents the priors connected to the data. Fig. 1(a) graphically gives the pipeline of the encoding-query process associated with (4) and (5). Following the framework, we could design a branchless neural network to form a spectral–spatial pansharpening INR, which we call NaivePINR.

To further strengthen the pansharpening, dynamic processing can be introduced, where the learn-to-learn tactic is utilized to produce adaptive adjustment [55]. The related training process is as follows:

$$\begin{aligned}&\boldsymbol{\theta}_g^{\text{opt}}, \boldsymbol{\theta}_s^{\text{opt}}, \boldsymbol{\theta}_r^{\text{opt}} \\ &= \arg\min_{\boldsymbol{\theta}_g, \boldsymbol{\theta}_s, \boldsymbol{\theta}_r} \mathcal{L}(\Phi(\boldsymbol{c}_t; \boldsymbol{r}_t = \Psi(\widetilde{\mathbf{H}}_t^{\uparrow}, \widetilde{\mathbf{P}}_t; \boldsymbol{\theta}_r), \boldsymbol{\theta}_g, \boldsymbol{\theta}_l = \varphi(\mathbf{S}_t; \boldsymbol{\theta}_s)), \mathbf{H}_t)\end{aligned} \tag{6}$$

where $\boldsymbol{\theta}_g$ denotes usual parameters directly related to processing data, $\boldsymbol{\theta}_l$ stands for dynamic parameters, and $\boldsymbol{\theta}_s$ represents learn-to-learn parameters of the dynamic branch $\varphi(\cdot)$, which yields those dynamic parameters. If dynamic parameters in $\boldsymbol{\theta}_l$ represent a learning rule adaptive to $\mathbf{S}_t$, which denotes the data input to the dynamic branch in the training stage, then learn-to-learn parameters in $\boldsymbol{\theta}_s$ are used to learn that learning rule. After obtaining $\boldsymbol{\theta}_g^{\text{opt}}$, $\boldsymbol{\theta}_s^{\text{opt}}$, and $\boldsymbol{\theta}_r^{\text{opt}}$ in the training stage, the subsequent inferring stage can be mathematically represented as

$$\widehat{\mathbf{H}}_a = \Phi(\boldsymbol{c}_f; \boldsymbol{r}_f = \Psi(\mathbf{H}^{\uparrow}, \mathbf{P}; \boldsymbol{\theta}_r^{\text{opt}}), \boldsymbol{\theta}_g^{\text{opt}}, \boldsymbol{\theta}_l' = \varphi(\mathbf{S}; \boldsymbol{\theta}_s^{\text{opt}})). \tag{7}$$

The optimal usual parameters and the optimal learn-to-learn parameters are determined in the training stage. When data $\mathbf{S}$ are input into the learn-to-learn branch, it would output dynamic parameters adaptive to those data. Then, the dynamic parameters and the optimal usual parameters are collaboratively used as a pansharpening rule to yield the pansharpening result. Fig. 1(b) shows the pipeline of a dynamic encoding–decoding process relevant to (6) and (7). Following the framework, we would develop a dynamic spectral–spatial pansharpening INR (DynamicPINR).

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Algorithm 1 NaivePINR for ARHS Pansharpening

Input:
HS data and PAN data for training stage and inferring stage $\mathbf{H}_t$, $\mathbf{P}_t$, $\mathbf{H}$ and $\mathbf{P}$; training scales $\boldsymbol{s}_t$ and the desired scale s_a for ARHS pansharpening

Output:
HS data on desired arbitrary scale $\widehat{\mathbf{H}}_a$;

Synthesize reduced-resolution HS and PAN images $\widetilde{\mathbf{H}}_t$ and $\widetilde{\mathbf{P}}_t$ by degrading with training scales $\boldsymbol{s}_t$;
Initialize model parameters $\boldsymbol{\theta}$ and $\boldsymbol{\theta}_r$ randomly;
for each epoch **do**
  Sample LR-HR batch for scale $s_{ai} \in \boldsymbol{s}_t$;
  Synthesize $\widetilde{\mathbf{H}}_t^{\uparrow\uparrow}$ by interpolating $\mathbf{H}_t$ with scale s_{ai};
  Obtain $\mathbf{H}_{\text{enc}}$ by spectral-spatial encoding;
  for (x, y) in spatial coordinates of $\boldsymbol{c}_t$ **do**
    Obtain $\boldsymbol{v}_L(x, y)$ by harmonic transform on (x, y);
    Obtain $\mathbf{H}_{\text{dec}}(x, y)$ with scale s' using Eq. (12);
  end for
  for b in spectral coordinates of $\boldsymbol{c}_t$ **do**
    Encode and broadcast b into $\boldsymbol{\Upsilon}_L(b)$ with the same spatial size as $\widetilde{\mathbf{H}}_t^{\uparrow\uparrow}$ sequentially;
    Obtain $\mathbf{H}_{\text{hyb}}(b)$ using Eq. (13);
    Obtain $\widehat{\mathbf{H}}_a(b)$ by spectral query;
  end for
  Calculate loss function $\mathcal{L}$;
  Update parameters $\boldsymbol{\theta}$ and $\boldsymbol{\theta}_r$ with respect to $\mathcal{L}$;
end for
Obtain optimal model parameters $\boldsymbol{\theta}^{opt}$ and $\boldsymbol{\theta}_r^{opt}$;
return $\widehat{\mathbf{H}}_a = \Phi(\boldsymbol{c}_f; \boldsymbol{r}_f = \Psi(\mathbf{H}^{\uparrow}, \mathbf{P}; \boldsymbol{\theta}_r^{opt}), \boldsymbol{\theta}^{opt})$;

IV. Naive Spectral–Spatial Pansharpening INR

INRs represent a technique to use parameterized neural networks to approximate continuous functions. This implies that those neural networks offer the potential to produce estimations on continuous coordinates, which could be utilized to resolve arbitrary-resolution enhancement. When we utilize the idea of INRs for ARHS pansharpening task, we have to develop specific INR networks that are capable of accommodating characteristics of HS data. In our design, we use a neural network to represent an HS scene as a 3-D continuous function. The function maps arbitrary spectral–spatial coordinates to the associated data, while the 3-D function is parameterized by the neural network. To generalize knowledge across various HS scenes while avoiding fitting individual implicit functions of them, the scene-specific prior (i.e., spectral–spatial latent code) needs to be extracted. We built our spectral–spatial INRs based on the encoding–decoding context, where spectral–spatial encoding is developed to refine the spectral–spatial latent codes. In the subsequent decoding phase, we established a spectral–spatial query mapping, which takes the spectral–spatial latent code as additional input accompanying the spectral–spatial coordinates and can be effective shared across all the possible scenes.

Fig. 2 graphically gives the overall architecture of our NaivePINR newly designed for ARHS pansharpening, where red dashed lines and connections are used for our DynamicPINR developed in later Section V. Basically, our NaivePINR is trunk only structured, comprising a spectral–spatial encoding to learn spectral–spatial latent space and a spectral–spatial query mapping procedure for decoding to reconstruct ARHS data.

Algorithm 2 DynamicPINR for ARHS Pansharpening

Input:
HS data and PAN data for training stage and inferring stage $\mathbf{H}_t$, $\mathbf{P}_t$, $\mathbf{H}$ and $\mathbf{P}$; training scales $\boldsymbol{s}_t$ and the desired scale s_a for ARHS pansharpening

Output:
HS image on desired arbitrary scale $\widehat{\mathbf{H}}_a$;

Synthesize the reduced-resolution HS images $\widetilde{\mathbf{H}}_t$ and the corresponding PAN images $\widetilde{\mathbf{P}}_t$ by degrading $\mathbf{H}_t$, $\mathbf{P}_t$ with training scales $\boldsymbol{s}_t$;
Initialize model parameters $\boldsymbol{\theta}_g$, $\boldsymbol{\theta}_s$, and $\boldsymbol{\theta}_r$ randomly;
for each epoch **do**
  Sample LR-HR batch for scale $s_{ai} \in \boldsymbol{s}_t$;
  Synthesize $\widetilde{\mathbf{H}}_t^{\uparrow\uparrow}$ by interpolating $\mathbf{H}_t$ with scale s_{ai};
  Obtain $\mathbf{H}_{\text{enc}}$ by spectral-spatial encoding;
  Obtain $\mathbf{H}_{\text{TST}}$ by TST module using Eq. (16);
  Obtain $\mathbf{H}_{\text{dy}}$ using Eq.(14);
  for (x, y) in spatial coordinates of $\boldsymbol{c}_t$ **do**
    Obtain $\boldsymbol{v}_L(x, y)$ by harmonic transform on (x, y);
    Obtain $\mathbf{H}_{\text{dec}}(x, y)$ with scale s' using Eq. (12);
  end for
  for b in spectral coordinates of $\boldsymbol{c}_t$ **do**
    Encode and broadcast b into $\boldsymbol{\Upsilon}_L(b)$ with the spatial size of $\widetilde{\mathbf{H}}_t^{\uparrow\uparrow}$ sequentially;
    Obtain $\mathbf{H}_{\text{hyb}}(b)$ using Eq. (13);
    for n in number of DyLinear layers N **do**
      Obtain parameters $\boldsymbol{\theta}_2'^n$ using Eq. (17);
      Obtain parameters $\boldsymbol{\theta}_2^n$ by low-rank composition;
      Calculate parameters $\boldsymbol{\theta}_3^n$ by element-wise product;
    end for
    Obtain $\widehat{\mathbf{H}}_a(b)$ using Eq. (18);
  end for
  Calculate loss function $\mathcal{L}$;
  Update parameters $\boldsymbol{\theta}_g$, $\boldsymbol{\theta}_s$, and $\boldsymbol{\theta}_r$ with respect to $\mathcal{L}$;
end for
Obtain optimal model parameters $\boldsymbol{\theta}_g^{opt}$, $\boldsymbol{\theta}_s^{opt}$, and $\boldsymbol{\theta}_r^{opt}$;
return $\widehat{\mathbf{H}}_a = \Phi(\boldsymbol{c}_f; \boldsymbol{r}_f = \Psi(\mathbf{H}^{\uparrow}, \mathbf{P}; \boldsymbol{\theta}_r^{opt}), \boldsymbol{\theta}_g^{opt}, \boldsymbol{\theta}_l' = \varphi(\mathbf{S}; \boldsymbol{\theta}_s^{opt}))$;

A. Spectral–Spatial Encoding

Various observed scenes comprise various ground objects and various spatial structures and thus are related to various spectral–spatial characteristics [3]. The specific spectral–spatial characteristic of a specific HS scene implies the prior of the connected HS data, i.e., the specific spectral–spatial latent code. We newly develop a spectral–spatial encoding process to learn the latent spectral–spatial space. On the one hand, the spectral–spatial encoding compacts the input data to strengthen the useful spectral–spatial information. On the other hand, the encoding yields a specific latent code in the

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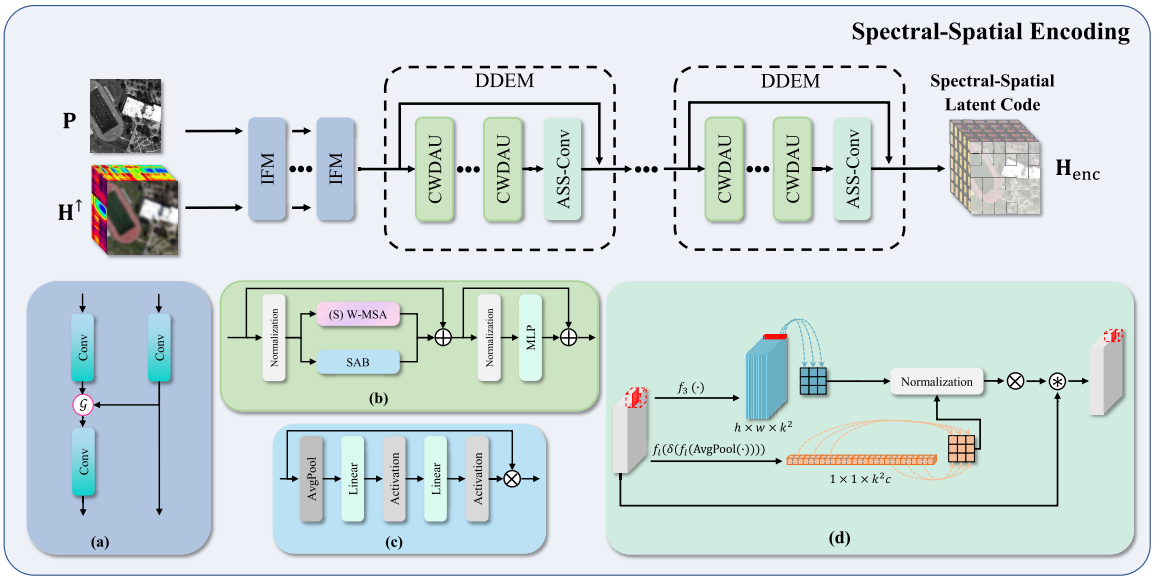


Fig. 3. Architecture of spectral–spatial encoding. ⊛ denotes the dynamic convolution. (a) IFM. (b) CWDAU. (c) SAB. (d) ASS-Conv.

spectral–spatial latent space highly adaptive to the input data, that is, when specific HS and connected PAN data are input to the spectral–spatial encoding, it could output a corresponding latent code, which can be used as the prior of the input data to collaboratively determine the relevant spectral–spatial pansharpening INR.

1) Architecture: Fig. 3 graphically illustrates the architecture of our designed spectral–spatial encoding in detail, which is in fact a dual-domain network. As shown in that figure, the process of spectral–spatial encoding consists of stacked initial fusion modules (IFMs) and several serialized dual-domain encoding modules (DDEMs). IFM is designed for the relatively sufficient information interaction of the pre-interpolated LRHS data $\mathbf{H}^{\uparrow}$ and the PAN data $\mathbf{P}$. It has two paths to enhance the features of two-source data separately and a gate recurrent unit (GRU) to fusion them. DDEMs build spectral–spatial dual-attention and spectral–spatial dual-adaptiveness to modulate data. Each of them is composed of some newly designed modules to compact the spectral–spatial information, including consecutive cross-window dual attention units (CWDAUs) and an adaptive spectral–spatial convolution (ASS-Conv). After the processing of the convolution layer and a series of DDEMs, the spectral–spatial latent code $\mathbf{H}_{\text{enc}}$ is obtained, which represents the prior of the specific input PAN and HS data.

2) Initial Fusion Module: As seen in Fig. 3(a), an IFM first uses two different convolution layers to process the HS deep features and PAN deep features. Then, a GRU is utilized to fulfill their information interaction. Besides, another convolution layer is adopted to further enhance the layer-wise fused features. For sufficient interaction of two-source data, we stack serval IFMs, which jointly build a PAN path and an HS path. Then, the output of the paths (i.e., HS and PAN layer-wise features) in the n_ith IFM can be mathematically represented as

$$\begin{cases} \mathbf{H}^{(n_i+1)} = f_3(\mathcal{G}(\mathbf{P}^{(n_i+1)}, f_3(\mathbf{H}^{(n_i)}))) \\ \mathbf{P}^{(n_i+1)} = f_3(\mathbf{P}^{(n_i)}) \end{cases} \tag{8}$$

where $\mathcal{G}(\cdot,\cdot)$ and $f_3(\cdot)$ denote the GRU and the 3×3 convolution, respectively, and $\mathbf{H}^{(n_i+1)}$ and $\mathbf{P}^{(n_i+1)}$ stand for the output of the n_ith IFM. It is noted that $\mathbf{H}^{(1)} = \mathbf{H}^{\uparrow}$ and $\mathbf{P}^{(1)} = \mathbf{P}$.

3) Consecutive Cross-Window Dual Attention Unit: As seen in Fig. 3(b), CWDAU is comprised of (shifted) window multihead self-attention [(S)W-MSA] [56], spectral attention block (SAB), MLP block, and some normalization layers. (S)W-MSA and SAB are parallel plugged with each other and then collaborate with a skip connection to form residual spectral–spatial attention operation to extract crucial spectral–spatial information. Therein, the input of SAB is sequentially processed by a spatial average-pooling, linear layers, and activation functions, and then, element-wise product multiplies a skip connection to produce its spectral attention feature [as shown in Fig. 3(c)]. Then, MLP and another skip connection are jointly used to fulfill a further feature enhancement for CWDAU. The process of CWDAU can be formulated as follows:

$$\begin{cases} \mathbf{X}_1 = \text{(S)W-MSA}(|\mathbf{X}_0|) + \text{SAB}(|\mathbf{X}_0|) + \mathbf{X}_0 \\ \mathbf{X}_2 = f_{\text{MLP}}(|\mathbf{X}_1|) + \mathbf{X}_1 \end{cases} \tag{9}$$

where $f_{\text{MLP}}(\cdot)$ denotes an MLP. For the sake of simplicity, we, respectively, denote the input and the output of the module being described as $\mathbf{X}_0 \in \mathbb{R}^{h\times w\times c}$ and $\mathbf{X}$ with the maximum subscript uniformly. $\mathbf{X}$ is closer to the output, as its subscript increases, where $|\cdot|$ denotes the normalization operation. The specific normalization is determined by its input, such as when $|\cdot|$ is applied to feature maps or kernels, it specifically refers to layer normalization or kernel normalization.

4) Adaptive Spectral–Spatial Convolution: As seen in Fig. 3(d), two paths are decoupled to produce dual-domain dynamic kernels in ASS-Conv. Specifically, its input is sequentially processed by spatial average-pooling, linear layers, an activation function in the lower path to produce spectral kernels $\mathbf{W}_{\text{spe}} \in \mathbb{R}^{1\times 1\times k^2 c}$, while in the upper path, a single convolution is used to produce spatial kernels $\mathbf{W}_{\text{spa}} \in \mathbb{R}^{h\times w\times k^2}$.

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Dual-domain kernels $\mathbf{W}_{\text{spe}}$ and $\mathbf{W}_{\text{spa}}$ can be formulated as

$$\begin{cases} \mathbf{W}_{\text{spe}} = f_l(\delta(f_l(\text{AvgPool}(\mathbf{X}_0)))) \\ \mathbf{W}_{\text{spa}} = f_3(\mathbf{X}_0) \end{cases} \tag{10}$$

where $f_l(\cdot)$ and $\delta(\cdot)$ denote the linear layer and the activation function, respectively. Then, spectral kernels and spatial kernels are merged by involving normalization and element-wise product operation to build a spectral–spatial kernel, which is spectral–spatial varying. Finally, the spectral–spatial varying kernels are used to process input data to strengthen spectral–spatial latent codes to their corresponding scenes. Thus, the enhancement result from ASS-Conv in the jth band of the ith pixel can be mathematically represented as

$$\mathbf{X}_1(i, j) = \mathbf{X}_0(\Omega_k(i), j) \otimes |\mathbf{W}_{\text{spa}}(i)| \otimes |\mathbf{W}_{\text{spe}}(j)| \tag{11}$$

where $\Omega_k(i)$ denotes the $k \times k$ window around the ith pixel (i.e., the ith window) and $\mathbf{W}_{\text{spa}}(i) \in \mathbb{R}^{1\times 1\times k^2}$ denotes the weight at the ith pixel in $\mathbf{W}_{\text{spa}}$. $\mathbf{W}_{\text{spe}}(j) \in \mathbb{R}^{1\times 1\times k^2}$ stands for the weight consisting of all jth features in each group of $\mathbf{W}_{\text{spe}}$, which is equally divided into k^2 groups.

B. Spectral–Spatial Query Mapping

In this section, we develop a spectral–spatial query mapping, which decodes the latent codes to establish a representation of HRHS images in a spectral–spatial continuous manner and thus reconstruct ARHS data. For this mapping, an LRHS scene is represented with a spectral–spatial latent code (i.e., its prior). If a spectral–spatial coordinate (a spatial position on a band) in the high-resolution domain is given, the mapping would use the coordinate to query the spectral–spatial latent code to output the corresponding HS value. As a coordinate could be continuous, the spectral–spatial query mapping builds a decoding that continuously represents HS data and therefore is able to produce HS images in desired arbitrary resolutions. As shown in Fig. 4, our newly designed spectral–spatial query mapping contains two basic processes, i.e., a spatial query process and a spectral query process.

1) Spatial Query: As shown in Fig. 4, to establish spatial query, spatial coordinates and an intermediate scale need to be collaborated with a spectral–spatial latent code via MLP to directly form the relationship between HS data and spatial coordinates. Spatial harmonic transform $\boldsymbol{\upsilon}_L(x, y) = [\sin(2^0 x), \cos(2^0 x), \sin(2^0 y), \cos(2^0 y), \ldots, \sin(2^{L-1} x), \cos(2^{L-1} x), \sin(2^{L-1} y), \cos(2^{L-1} y)]$ could be used band by band, where L is the number of sine or cosine harmonics. Through the coordinate transform, the coordinate (x, y) is transferred to $\boldsymbol{\upsilon}_L(x, y)$.

As shown in Fig. 4, by using a given spectral–spatial latent code, the desired transformed coordinates, and a desired zoom scale, MLP first produces an initial query result. Subsequently, the result further performs an area weighting operation to achieve a local ensemble, which is related to a consideration of spatial dependence. Thus, the final result of spatial query can be mathematically represented as

$$\mathbf{H}_{\text{dec}}(x, y) = \sum_{\omega \in \Omega_2'(x,y)} \frac{A_\omega}{A} f_{\text{MLP}}(\mathbf{z}_\omega^*, \text{Cat}(s', \boldsymbol{\upsilon}_L(x, y))) \tag{12}$$

where $\Omega_2'(x, y)$ and ω denote the 2×2 window around (x, y) and each pixel element in it, respectively. A_ω denotes the area of the rectangle between (x, y) and ω, while $A = \sum_\omega A_\omega$ denotes the total area of $\Omega_2'(x, y)$. $\mathbf{z}_\omega^*$ denotes a pixel of latent code at ω from implicit dual-domain representation space. $s' = s_a/s$ denotes the intermediate scale between the ARHS pansharpened image and the PAN image. $\text{Cat}(\cdot, \cdot)$ represents the concatenation operation.

2) Spectral Query: When performing spectral decoding, the model should take the coordinate of query band b and the result from the previous stage $\mathbf{H}_{\text{dec}}$ into consideration. As mentioned in Section IV-B1, we still utilize the harmonic transform to keep the ability of modeling high-frequency details. The harmonic transform result is $\boldsymbol{\Upsilon}_L$. Besides, it is noteworthy that the interpolated version of $\mathbf{H}^{\uparrow}$ with s' (i.e., $\mathbf{H}^{\uparrow\uparrow}$) can be used as a useful guidance feature to promote the spectral query performance. For this reason, we are expected to feed the bth band of $\mathbf{H}^{\uparrow\uparrow}$ [i.e., $\mathbf{H}^{\uparrow\uparrow}(b)$] into the spectral query during the practical fulfillment. Therefore, we design a hybrid feature $\mathbf{H}_{\text{hyb}}$, which integrates $\boldsymbol{\Upsilon}_L$, $\mathbf{H}^{\uparrow\uparrow}(b)$, and $\mathbf{H}_{\text{dec}}$. The formulation of this process in the bth band is as follows:

$$\mathbf{H}_{\text{hyb}}(b) = \text{Cat}(\boldsymbol{\Upsilon}_L(b), \mathbf{H}^{\uparrow\uparrow}(b), \mathbf{H}_{\text{dec}}). \tag{13}$$

Then, the bth band, which is spectral sampling from continuous bands, of reconstructed ARHS image $\widehat{\mathbf{H}}_a$ is obtained by the forward propagation of spectral MLP.

Our NaivePINR designed in this section can be summarized in Algorithm 1.

V. Twofold Dynamic Spectral–Spatial INR

In HS pansharpening tasks, the PAN and the HS data are usually acquired over natural scenes. Consequently, the characteristics of those PAN and HS data are often with spatial variation, which means that their characteristics vary spatially, and thus, the method design of HS pansharpening could take into consideration certain spatial adaptability to follow the spatial variation. Based on NaivePINR proposed in Section IV, here we developed DynamicPINR to deal with spatial variation. In DynamicPINR, spatial dynamics are considered in twofold: one is a dynamic spectral–spatial encoding to yield dynamic latent codes and the other is a dynamic query mapping to dynamically establish coordinate-data connections.

A. Dynamic Spectral–Spatial Encoding

To strengthen the spatial dependencies dropped in pixel-wise spatial decoding, we can adjust the rule of obtaining spectral–spatial latent codes to enforce global spatial relationship in those latent codes by virtue of the following consideration:

$$\mathbf{H}_{\text{dy}} = \mathbf{H}_{\text{enc}} + \varphi_{d1}(\boldsymbol{\theta}_{d1}) \tag{14}$$

where $\mathbf{H}_{\text{enc}}$ denotes the spectral–spatial latent codes mentioned in Section IV-B, $\mathbf{H}_{\text{dy}}$ represents dynamic spectral–spatial latent codes we design here, $\varphi_{d1}(\cdot)$ stands for a network branch to acquire global spatial dependence, and $\boldsymbol{\theta}_{d1}$ denotes the parameters of the branch. Fig. 5 graphically gives our

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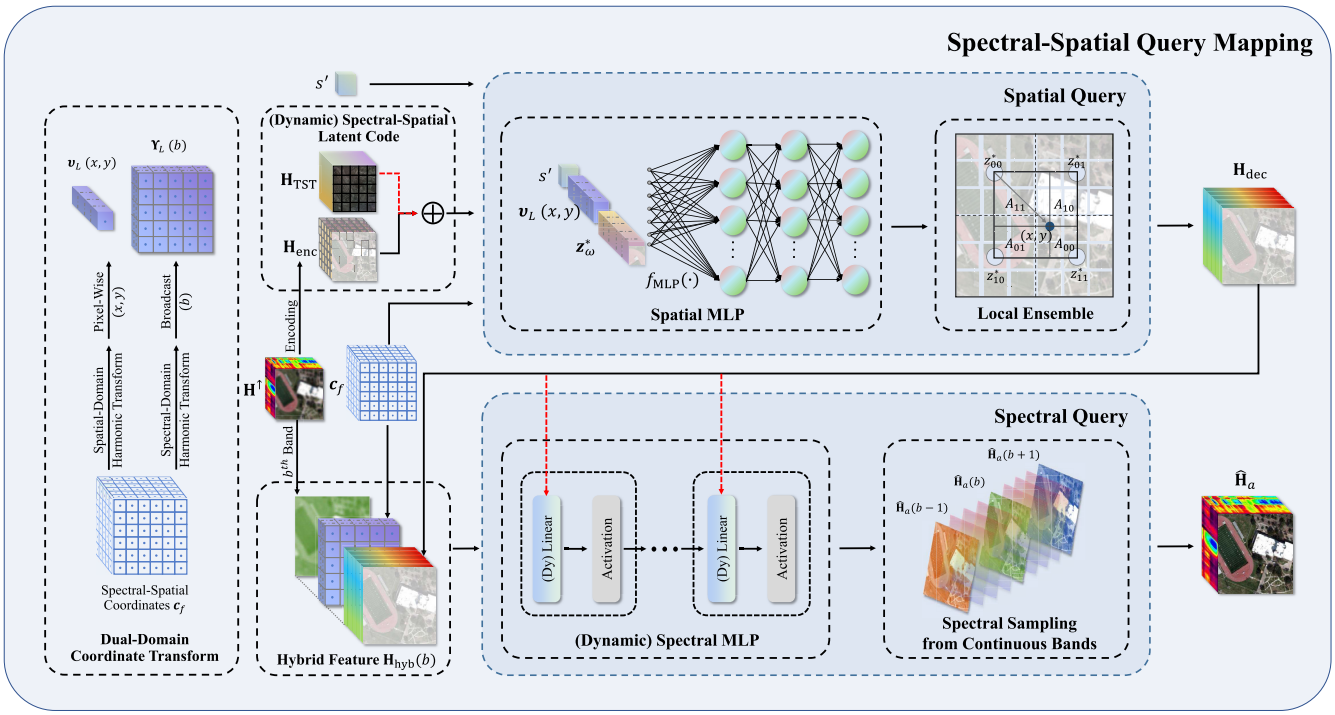


Fig. 4. Architecture of spectral–spatial query mapping.

TABLE I
PROPERTIES OF DATASETS USED FOR EVALUATION

Datasets	Spectral Range	Training HS	Training PAN	Inferring HS	Inferring PAN	Instrument
Bologna	0.4-2.5 μm	HS 400*200*69	PAN 2400*1200	HS 33*33*69	PAN 198*198	PRISMA
Changping	0.4-2.5 μm	HS 1200*180*137	PAN 3600*540	HS 60*60*137	PAN 180*180	Hyperion
Chikusei	0.4-1.0 μm	HS 100*100*124	PAN 500*500	HS 36*36*124	PAN 180*180	Headwall Hyperspec-VNIR-C
Pavia Center	0.4-0.8 μm	HS 200*40*102	PAN 1000*200	HS 50*50*102	PAN 200*200	ROSIS

designed instance of $\varphi_{d1}(\boldsymbol{\theta}_{d1})$, which is a TST module. More texturally, similar features in global spatial dependencies [57] are expected to be found by our TST module. Therefore, $\mathbf{H}_{\mathrm{TST}} = \varphi_{d1}(\boldsymbol{\theta}_{d1})$ stands for TST features, which are associated with global spatial dependencies.

As shown in Fig. 5, our innovative TST first uses texture feature extraction (TFE) blocks, each of which consists of a single convolution layer and consecutive residual blocks, to transform PAN images and HS images to their feature spaces. Those features are utilized as queries and keys of our newly designed texture similarity embedding (TSE) operation and intermediate features. The structure of TFE is shown in Fig. 5(a). The TSE operation is then built to find long-range dependence-based textural similarities between the PAN source and the HS source. The structure of TSE is shown in Fig. 5(b). In TSE, two query–key pairs (i.e., $\mathbf{Q}_{\boldsymbol{\gamma}}$–$\mathbf{K}_{\boldsymbol{\gamma}}$ and $\mathbf{Q}_{\boldsymbol{\beta}}$–$\mathbf{K}_{\boldsymbol{\beta}}$) output by TFE blocks are fed into a processing of nonlocal band-wise matching to calculate global texture similarities between two sources, which are connected to the coefficients $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ used in the final processing in TST [i.e., texture affine transform (TAT)]. It is noteworthy that there is a slight difference between two TSE operations used to generate $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$. More specifically, one TSE with kernel normalization [TSE (w/ KN)] produces the embedding of $\mathbf{Q}_{\boldsymbol{\gamma}}$-$\mathbf{K}_{\boldsymbol{\gamma}}$, while the other TSE without kernel normalization [TSE (w/o KN)] yields the embedding of $\mathbf{Q}_{\boldsymbol{\beta}}$-$\mathbf{K}_{\boldsymbol{\beta}}$. Through the nonlocal band-wise matching, the value of $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ in the jth band at spatial position (i_1, i_2) can be mathematically represented as

$$\begin{cases} \boldsymbol{\gamma}(i_1, i_2, j) = \langle |\mathbf{Q}_{\boldsymbol{\gamma}}(\Omega_3(i_1), j)|, |\mathbf{K}_{\boldsymbol{\gamma}}(\Omega_3(i_2), j)| \rangle \\ \boldsymbol{\beta}(i_1, i_2, j) = \langle \mathbf{Q}_{\boldsymbol{\beta}}(\Omega_3(i_1), j), \mathbf{K}_{\boldsymbol{\beta}}(\Omega_3(i_2), j) \rangle \end{cases} \tag{15}$$

where $\langle \cdot, \cdot \rangle$ stands for matrix inner product. Finally, we establish the following TAT:

$$\mathbf{H}_{\mathrm{TST}} = \boldsymbol{\gamma} \otimes \mathbf{V} + \boldsymbol{\beta} \tag{16}$$

to fully tune intermediate data $\boldsymbol{V}$ with global texture similarities $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$, thus yielding TST features $\mathbf{H}_{\mathrm{TST}}$.

In TAT, the affine coefficients $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ are dynamically adaptive to the input PAN and LRHS, and thus, the parameters $\boldsymbol{\theta}_{d1}$ of the TST branch $\varphi_{d1}(\boldsymbol{\theta}_{d1})$ are dynamically related to the input data. Once $\boldsymbol{\theta}_{d1}$ is determined, $\varphi_{d1}(\cdot)$ outputs the features adaptive to certain HS data and dynamically adjusting the rule of yielding spectral–spatial encoding. Therefore, we have the dynamic spectral–spatial latent code $\mathbf{H}_{\mathrm{dy}}$.

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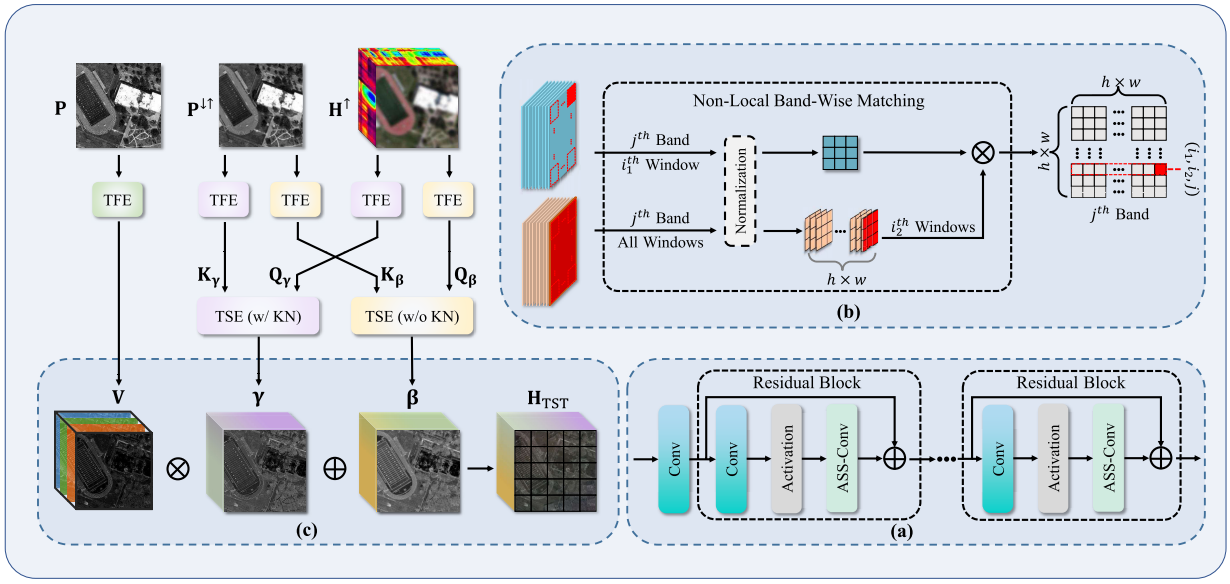


Fig. 5. Architecture of TST module. (a) TFE. (b) TSE. (c) TAT.

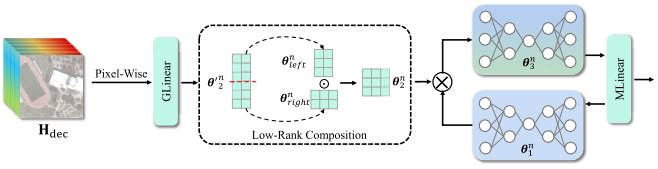


Fig. 6. Block diagram of a DyLinear layer in the spectral MLP.

B. Dynamic Spectral–Spatial Query Mapping

Our dynamic spectral–spatial query mapping introduces the consideration of data-adaptive query processing. In our naive spectral–spatial INR, spectral–spatial query mapping is composed of both spectral query and spatial query. A spectral query is crucial for a spectral recovery of the final pansharpened HS data. To further strengthen the spectral preserving, the result of the spatial query can be used to adjust the spectral query. As shown in Fig. 4, we design a novel dynamic spectral query initiated by spatial query, where the result of spatial query is used to modulate the parameters of MLP in the spectral query using a low-rank composition, thus forming a learn-to-learn dynamic [55] subnetwork.

However, using the learn-to-learn branch to directly tune the parameters of MLP is extremely inefficient due to the large number of parameters in the practically used MLP. This may lead to infeasible computation in the practical tuning of MLP. Here, we adopt a low-rank composition trick to deal with the problem. Fig. 6 graphically gives the basic thread of the trick. In the spectral MLP dynamically tuned, dynamic linear (DyLinear) layers are dynamically adjusted by the spatial query result. As shown in Fig. 6, each DyLinear layer is composed of a generating linear (GLinear) layer and a modulated linear (MLinear) layer. In the nth DyLinear layer, a GLinear layer first utilizes the spatial query result $\mathbf{H}_{\text{dec}}$ to generate temporary parameters $\boldsymbol{\theta}_2'^n \in \mathbb{R}^{(n_{\text{in}}^n + n_{\text{out}}^n)\times e}$ as

$$\boldsymbol{\theta}_2'^n = f_{gl}^n(\mathbf{H}_{\text{dec}}) \tag{17}$$

where $f_{gl}^n(\cdot)$ denotes the GLinear layer in the nth DyLinear layer, e denotes the given low rank, and n_{in}^n and n_{out}^n represent the numbers of input neural nodes and output neural nodes of the nth DyLinear layer, respectively. Then, we split temporary parameters $\boldsymbol{\theta}_2'^n$ into $\boldsymbol{\theta}_{\text{left}}^n \in \mathbb{R}^{n_{\text{in}}^n \times e}$ and $\boldsymbol{\theta}_{\text{right}}^n \in \mathbb{R}^{e \times n_{\text{out}}^n}$ and further perform matrix multiplication on them to calculate the modulating parameters $\boldsymbol{\theta}_2^n \in \mathbb{R}^{n_{\text{in}}^n \times n_{\text{out}}^n}$ (i.e., $\boldsymbol{\theta}_2^n = \boldsymbol{\theta}_{\text{left}}^n \odot \boldsymbol{\theta}_{\text{right}}^n$, where $\odot$ denotes the matrix multiplication). The static parameters (i.e., the frozen parameters during the inferring stage) of the MLinear layer in the nth DyLinear $\boldsymbol{\theta}_1^n \in \mathbb{R}^{n_{\text{in}}^n \times n_{\text{out}}^n}$ is modulated by $\boldsymbol{\theta}_2^n$ using an element-wise product to obtain $\boldsymbol{\theta}_3^n \in \mathbb{R}^{n_{\text{in}}^n \times n_{\text{out}}^n}$, following $\boldsymbol{\theta}_3^n = \boldsymbol{\theta}_1^n \otimes \boldsymbol{\theta}_2^n$. Then, the bth band of reconstructed ARHS image $\widehat{\mathbf{H}}_a$ sampling from continuous bands is obtained as follows:

$$\widehat{\mathbf{H}}_a(b) = f_{\text{ml-}\delta}^N(\ldots(f_{\text{ml-}\delta}^1(\mathbf{H}_{\text{hyb}}(b); \boldsymbol{\theta}_3^1))\ldots; \boldsymbol{\theta}_3^N) \tag{18}$$

where $f_{\text{ml-}\delta}^n(\cdot)$ refers to the mapping function implemented by an MLinear layer and an activation function in the nth DyLinear layer and N is the number of DyLinear layers in spectral query.

Our DynamicPINR developed in this section offers a way to deal with the ARHS pansharpening task by combining our spectral–spatial INR with elaborate dynamic considerations. Its practical implementation is summarized in Algorithm 2.

VI. Experimental Results

In this section, we report the experiments used to validate the performances of our designed DynamicPINR and NaivePINR, including experimental datasets, quantitative metrics, implementation details, ARHS pansharpening experiments, and standard HS pansharpening experiments.

A. Experimental Datasets

In this section, our methods are tested on four datasets, including the Bologna dataset, the Changping dataset, the

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Chikusei dataset, and the Pavia Center dataset. Bologna and Changping are as real datasets in our experiments, which contain the real HS images and the connected PANs, whereas Chikusei and Pavia Center are as simulated datasets, which comprise the real HS images but no PAN images.

1) *Bologna Dataset:* The HS data and the PAN data in the dataset are simultaneously acquired over the area of Bologna city, Italy, using the Hyperspectral Precursor of the Application Mission (PRISMA) satellite. The HS data comprising 234 bands cover a spatial size of 400×400 with the spatial resolution of 30 m, spectrally ranging from 400 to 2500 nm. The PAN image is with a spatial covering of 2400×2400 and a spatial resolution of 5 m. An operation of recalibration and noisy-band removal was used to select 69 bands.
2) *Changping Dataset:* This dataset is captured over Changping District, Beijing, China, by using the Hyperion spectrometer and the Advanced Land Imager (ALI) on the Earth Observing One (EO-1) satellite. The HS image is with 242 bands spectrally covering from 400 to 2500 nm with a spatial scope of 2900×180 where the spatial resolution is 30 m, and the PAN image is with spatial size of 8700×540 and a spatial resolution of 10 m. With an operation of recalibration and noisy-band removal, 137 bands were remained.
3) *Chikusei Dataset:* The HS data of this dataset are with a spatial range of 2517×2335 pixels, with a spectral range from 363 to 1018 nm, which covers 128 bands. The Headwall Hyperspec-VNIR-C imaging sensor takes the dataset over the Chikusei area, Ibaraki, Japan. In our experiments, four noisy and polluted bands were deleted.
4) *Pavia Center Dataset:* Pavia Center was taken by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over an area of Pavia, Italy, comprising HS data with a spatial size of 1096×715 and with 115 bands covering a spectral range of 430–860 nm. Our experiments discarded 13 noisy and polluted bands.

B. Compared Methods and Quantitative Metrics

Our experiments use ten existing HS pansharpening methods to perform a comparison with two methods we develop in this work, i.e., NaivePINR and DynamicPINR. Five of those ten existing methods are representative classical methods, which are GSA [8], SFIM [13], CNMF [19], MTF-GLP [14], and BayesSparse [18]. Three of them are CNNs, which are built for standard HS pansharpening task, including PNN [20], HyperPNN [28], and HSpeNet [29]. To enable these eight methods, which are originally used for standard pansharpening task, to perform ARHS pansharpening task, we add an operation of bicubic interpolation as the last operating step of those methods, wherein three modified CNN methods are therefore called PNN-Bic, HyperPNN-Bic, and HSpeNet-Bic hereafter. Furthermore, we also include two recently developed ARHS pansharpening approaches, i.e., ARHS-CNN [36] and VSPC-CNN [37], in experiments. In VSPC-CNN [37], DARN [58] is used as the prepansharpening subnetwork. Therefore, there are totally 12 methods performed in our experimental comparison.

Our experiments include ARHS pansharpening experiments and standard HS pansharpening experiments. The former ones are used to test our methods' abilities to pansharpen HS data to any desired resolutions; the latter ones aim to verify that our ARHS pansharpening methods still have high capabilities to pansharpen HS data to the resolutions of connected PANs. In each of the two types of pansharpening experiments, pansharpening methods are evaluated both via reduced-resolution assessment (i.e., at the scale of the synthetic reduced-resolution LRHS) and via full-resolution assessment (i.e., at the scale of the observed LRHS) [36], [37], [59]. Reduced-resolution assessment considers quality indexes between the observed LRHS and the pansharpening result from the synthetic reduced-resolution LRHS and PAN [36], [37], [60]. Four widely used quality indexes are adopted, including cross correlation (CC), spectral angle mapper (SAM), root-mean-square error (RMSE), and peak signal-to-noise ratio (PSNR). Full-resolution assessment realizes pansharpening evaluation directly between the observed HS and PAN data and the pansharpening result from those observed data. More specifically, the assessment is performed using the quality with on reference (QNR), the spatial distortion index D_s, and the spectral distortion index D_λ [61].

C. Implementation Details

In this section, we mainly introduce the key implementation details, including hyperparameters of the proposed models and experiments. There are three parts (i.e., spectral–spatial encoding, spectral–spatial query mapping, and TST module) of hyperparameters settings mainly illustrated. For spectral–spatial encoding, we set the numbers of IFMs and DDEMs are 1 and 3, respectively. The number of CWDAUs in each DDEM is [2, 4, 2]. The reduced ratio of spectral dimension in SAB is set to 16, and all output dimensions of those layers in the encoding are 64. For spectral–spatial query mapping, the number of sine or cosine harmonics is empirically set to 5, and the output dimensions of the linear layers in spatial query and spectral query are [256, 256, 256, 64] and [256, 256, 256, 1], respectively. Rank e is empirically set to 3. For the TST module, the numbers of residual blocks in TFEs are all set to 1. The activation function in backend of SAB is the sigmoid function, while the others are the leaky rectified linear activations with 0.2 slope on negative inputs. k (i.e., the window size in ASS-Conv) is set to 3. It is noteworthy that all spectral–spatial coordinates are normalized in the range $[-1, 1]$ along the spatial dimension and the spectral dimension.

Each dataset used in our experiments is formed into three partitions, including the training, the validation, and the testing regions. For the training stage, the training region and the validation region are degraded according to the given training scales and then divided them into patches. During training, the setting of hyperparameters, such as learning rate, is driven by the pansharpening result on the validation region. Then, the testing region is used for the inferring stage to perform the reduced-resolution assessment and the full-resolution assessment.

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In the reduced-resolution assessment, we treat the observed LRHS image as the reference/ground truth (GT). Then, we need to degrade the observed LRHS data and PAN data to produce a reduced-resolution version of the PAN and the LRHS data as the input of a pansharpening processing. The following gives those details of the data preprocessing to yield the reduced-resolution LRHS data and the reduced-resolution PAN data. For the standard HS pansharpening, we synthesize the LRHS image that needs to be pansharpened and the connected PAN using those observed HS and PAN data. More specifically, the reduced-resolution LRHS image and the connected reduced-resolution PAN image are produced with Gaussian blurring and downsampling according to the pansharpening scale. For the two datasets Bologna and Changping, there are PAN images available and the above reduced-resolution operations can be directly conducted on both the observed PAN and HS data. For the other two datasets, i.e., Chikusei and Pavia Center, the real observed PAN image of each of the datasets is unavailable. Therefore, the reduced-resolution PAN is synthesized with the average over all the visible bands of the observed LRHS data. For the ARHS pansharpening, we also use Gaussian blurring and downsampling to synthesize a series of reduced-resolution LRHS images and the connected reduced-resolution PANs according to the given pansharpening scales $\{s_{ai}\}(i = 1, 2, 3, \ldots, N_s)$ that set for training (a pansharpening scale s_{ai} corresponding to an intermediate scale s_i'). For datasets Bologna and Changping, the reduced-resolution operations are directly performed on both the observed PAN and HS data. For datasets Chikusei and Pavia Center, the real observed PANs are absent and the reduced-resolution data producing comprises two steps. In the first step, the observed HS data are degraded according to the intermediate scale $s_i' = s_{ai}/s$, and then, the reduced-resolution PAN are synthesized with the average over all the visible bands of the degraded HS data. In the second step, the degraded HS data are further degraded in terms of the scale s, thus yielding the final reduced-resolution HS data. In addition, as our naivePINR and DynamicPINR need a pre-interpolation for their inputs, we use bicubic interpolation with scale $s = s_{ai}/s_i'$ to interpolate LRHS to the spatial size of PAN and then combine them as a low-resolution (LR) pair. More specifically, we extract LR patches with the spatial size of 20×20 randomly from the training region, and each of them is used as a training sample.

Our experiments are run on an NVIDIA GeForce GTX 3060 GPU and by Pytorch framework, with Inter Core i7-10700F CPU. Classical methods are conducted by the codes of the publication [2], while CNN-based methods, i.e., PNN, HyperPNN, HSpeNet, ARHS-CNN, and VSPC-CNN, are implemented strictly following the description details of their associated works. For relatively convincing comparisons, all learning-based methods, i.e., NaivePINR, DynamicPINR, PNN-Bic, HyperPNN-Bic, HSpeNet-Bic, ARHS-CNN, and VSPC-CNN, are uniformly optimized in 300 epochs by the same optimizer AdamW [62], [63] with weight decay 10^{-4} and initial learning rate 5×10^{-4}. Besides, we select a set of s_i' (from 1.0 to 2.5 with a stride of 0.1) and set the number of training samples (i.e., 20×20 LR patches and their references/GTs) for each scale to 256, and thus, the total number of them is 4096. If necessary, we perform augmentation of the training patches by the operation of random flipping and rotating. The descriptions of the datasets used in our experiments and their partitioning usages are given in Table I.

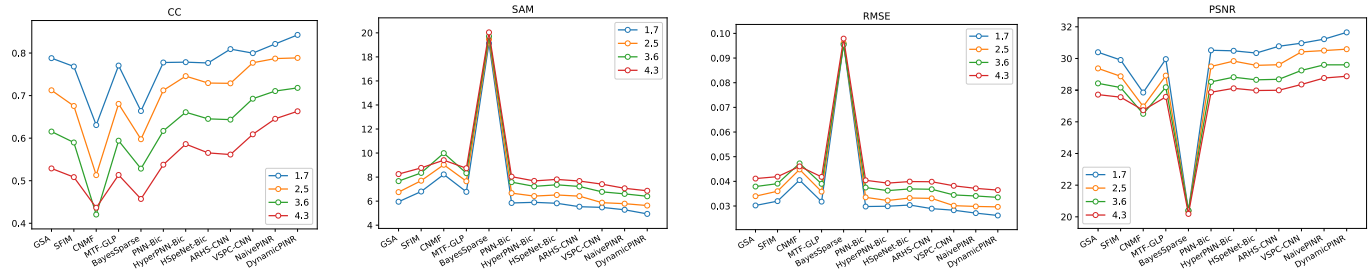
D. ARHS Pansharpening Experiments

ARHS pansharpening experiments are conducted to test the abilities of our NaivePINR and DynamicPINR to pansharpen LRHS data to any spatial resolution.

1) Reduced-Resolution Assessment: We first test the ARHS pansharpening performances of our methods at the reduced-resolution level. In the experiments, we randomly select $s' = 1.7, 2.5, 3.6,$ and 4.3 and $s' = 1.5, 1.7, 2.5, 3.2, 4.3, 4.7, 5.2,$ and 6.1 as intermediate scales for the four datasets to perform quantitative comparisons and visual evaluations. For quantitative assessments, we first use the observed LRHS image as the reference/GT and then degrade the observed LRHS and PAN images with intermediate scales. Those synthetic reduced-resolution data are used as the inputs of a pansharpening method. Then, the output of the method is assessed by reduced-resolution quality metrics. Figs. 7–10 illustrate the quantitative results of ARHS pansharpening on Bologna, Changping, Chikusei, and Pavia Center datasets, respectively, and Figs. 11–14 display their graphical results. As can be seen, our NaivePINR and DynamicPINR outperform the other methods in CC, SAM, RMSE, and PSNR for all the four resolutions, especially when the scale is relatively large. Moreover, the dynamic designs in DynamicPINR jointly contribute to further boosting the ARHS pansharpening performance. In addition, interpolation-attached methods, including GSA, SFIM, CNMF, MTF-GLP, BayesSparse, PNN-Bic, HyperPNN-Bic, and HSpeNet-Bic, yield worse performance than ARHS pansharpening algorithms, as the former ones utilize bicubic interpolation to introduce possible useless textures or details to pansharpened ARHS images, while the latter ones use specific scale generalization mechanism to relatively fully refine the data structure to achieve ARHS zoom. Also, it could be observed that those learning-based methods outperform those representative classical methods, which can be attributed to the adaptability of learnable parameters to various dataset characteristics. It also deserves a notice that CC, SAM, RMSE, and PSNR tend to drop with the increase of test scale s'. Such a phenomenon and an intuition coincide, which is that the higher target spatial resolution results in harder ARHS pansharpening.

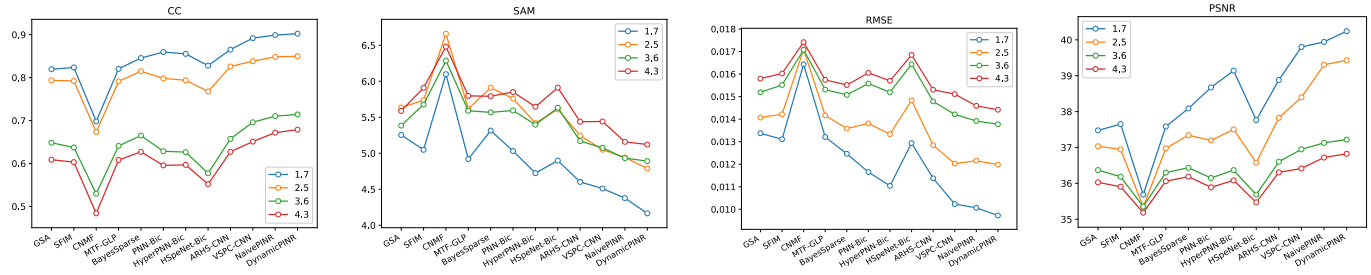
Except for quantitative evaluation, we additionally perform a visual inspection and hence qualitative evaluation. In the training stage, the reduced-resolution data under a set of s_i', which are from 1.0 to 2.5 with a stride of 0.1, are used. Then, in the inferring stage, the DynamicPINR network with trained parameters inputs reduced-resolution LRHS and PAN images degraded at a single scale and outputs various ARHS pansharpening results connected to the given ARHS pansharpening scales. We present the visualizations of the ARHS pansharpening results of our DynamicPINR in Figs. 11–14, involving pansharpening scales beyond those in the training stage, where the intermediate scale s' is set to be 1.5, 1.7, 2.5,

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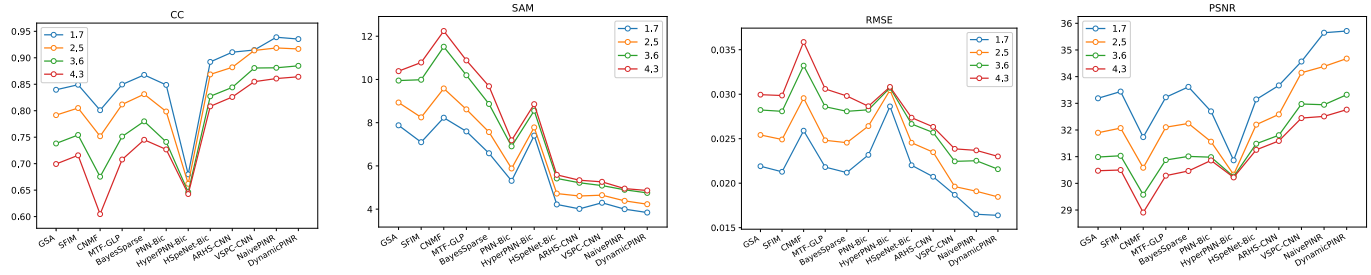
(a) (b) (c) (d)

Fig. 7. Pansharpening performances under arbitrary scales on the Bologna dataset. (a) CC. (b) SAM. (c) RMSE. (d) PSNR. Note that intermediate scales s' are given in the figure.



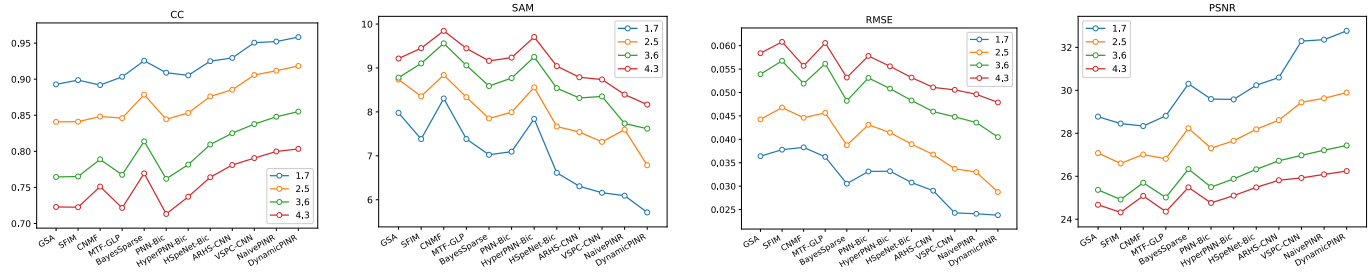
(a) (b) (c) (d)

Fig. 8. Pansharpening performances under arbitrary scales s_a on the Changping dataset. (a) CC. (b) SAM. (c) RMSE. (d) PSNR.



(a) (b) (c) (d)

Fig. 9. Pansharpening performances under arbitrary scales s_a on the Chikusei dataset. (a) CC. (b) SAM. (c) RMSE. (d) PSNR.



(a) (b) (c) (d)

Fig. 10. Pansharpening performances under arbitrary scales s_a on the Pavia Center dataset. (a) CC. (b) SAM. (c) RMSE. (d) PSNR.

3.2, 4.3, 4.7, 5.2, and 6.1. As observed in Figs. 11–14, with the growing pansharpening scale, the qualities of reconstructed HRHS images by our ARHS pansharpening methods do not incur obvious decline, especially at those relatively high scales.

2) Full-Resolution Assessment: Full-resolution assessment is used to further validate the generalization ability of our ARHS pansharpening methods on real observed HS and PAN data. All the compared methods, including GSA, SFIM, CNMF, MTF-GLP, BayesSparse, PNN-Bic, HyperPNN-Bic, HSpeNet-Bic, ARHS-CNN, VSPC-CNN, NaivePINR, and DynamicPINR, perform the ARHS pansharpening on the real HS and PAN data of Changping and Bologna datasets. Similar to the experiments for reduced-resolution assessment, those methods designed for standard pansharpening

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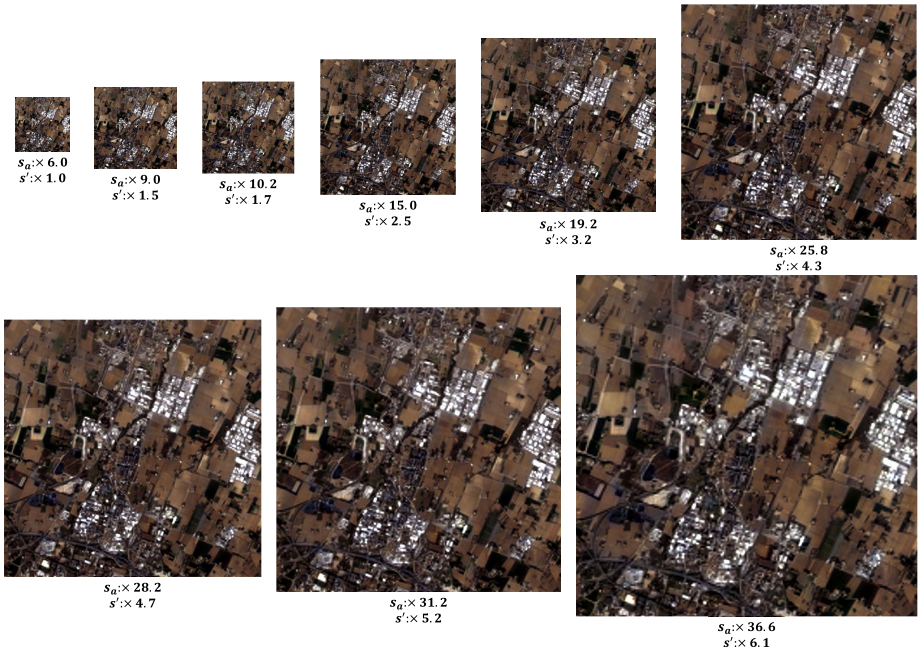


Fig. 11. Visualization of ARHS pansharpening by using DynamicPINR on the reduced-resolution Bologna dataset.

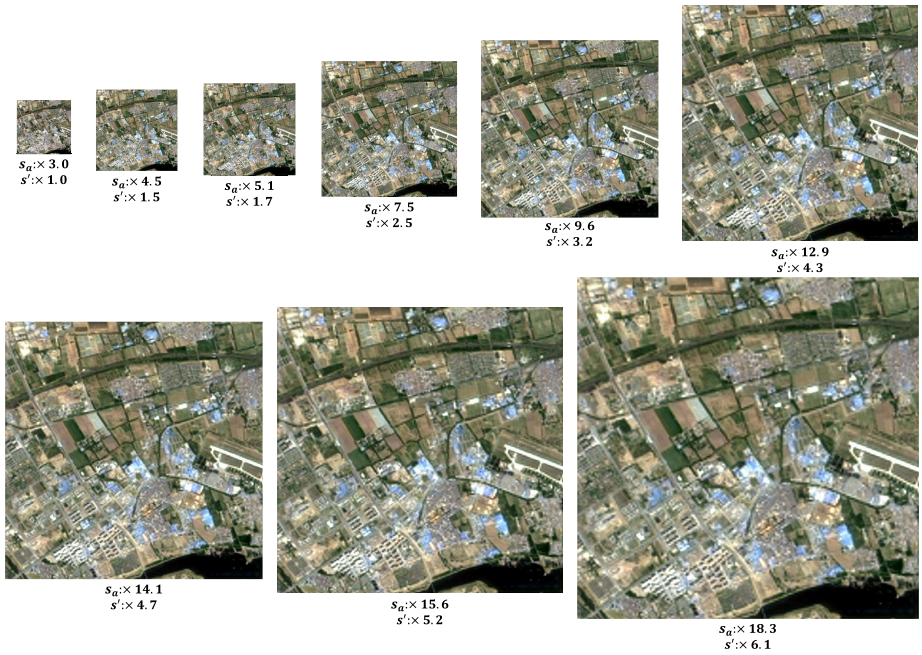


Fig. 12. Visualization of ARHS pansharpening by using DynamicPINR on the reduced-resolution Changping dataset.

are added an operation of bicubic interpolation as their final step to achieve the ARHS pansharpening task. For ARHS pansharpening-specific methods, including ARHS-CNN, VSPC-CNN, NaivePINR, and DynamicPINR, we set the intermediate scale for the inferring stage as $s' = 1.7, 3.6,$ and 5.2. For the Bologna dataset, the spatial resolution ratio between observed PAN and LRHS, i.e., s, is 6, and then, we have ARHS pansharpening scale $s_a = 10.2, 21.6,$ and 31.2. For Changping, $s = 3$, and then, we have ARHS pansharpening scale $s_a = 5.1, 10.8,$ and 15.6.

Fig. 15(a)–(l) graphically give the results of the above 12 methods on the Changping dataset. The upper part of each figure gives RGB images composed of three bands in the visible range, while the lower part shows false-color images synthesized by three bands out of the visible range. Fig. 16(a)–(l) presents the similar visual comparison on the Bologna dataset. As observed in Figs. 15 and 16, our NaivePINR and DynamicPINR successfully alleviate blurry textures and reconstruct more natural edges and high-frequency details than the other compared methods, and

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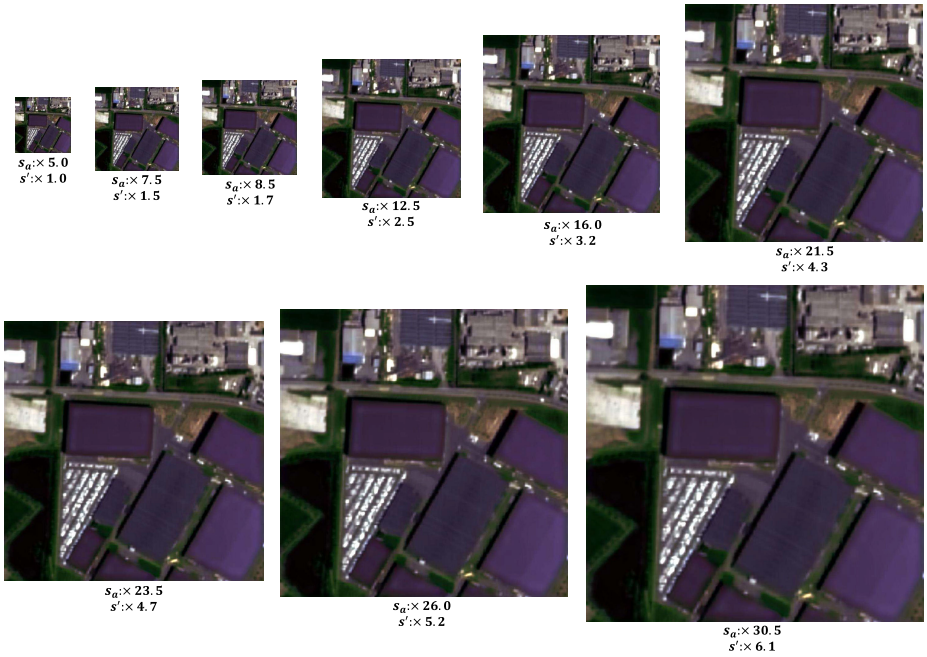


Fig. 13. Visualization of ARHS pansharpening by using DynamicPINR on the reduced-resolution Chikusei dataset.

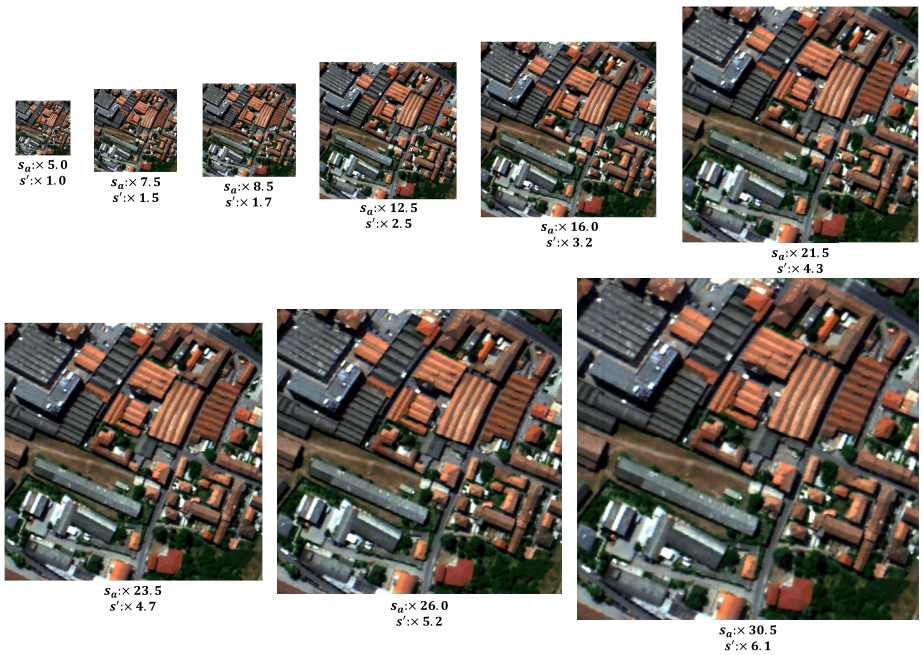


Fig. 14. Visualization of ARHS pansharpening by using DynamicPINR on the reduced-resolution Pavia Center dataset.

in particular, the latter one produces less incorrect color, which means that the twofold dynamic mechanism in DynamicPINR additionally helps to deal with spectral distortion.

E. Standard HS Pansharpening Experiments

We use experiments here to validate that our ARHS pansharpening methods still keep the high ability to fulfill standard HS pansharpening tasks.

1) Reduced-Resolution Assessment: It is worth noting that if the spatial resolution of target ARHS pansharpened data is set to be the same as that of the PAN, i.e., $s' = 1$ or $s_a = s$, we then degenerate ARHS pansharpening to standard HS pansharpening, that is, standard HS pansharpening is in fact a special case of ARHS pansharpening.

For all four datasets, we conducted standard HS pansharpening on the reduced-resolution data and then performed evaluations using quantitative indexes CC, SAM, RMSE, and PSNR. Tables II–V numerically give the standard HS pansharpening results. We use boldface and underline to mark the best and the second-best scored, respectively. As can be easily observed in Tables II–V, our DynamicPINR and

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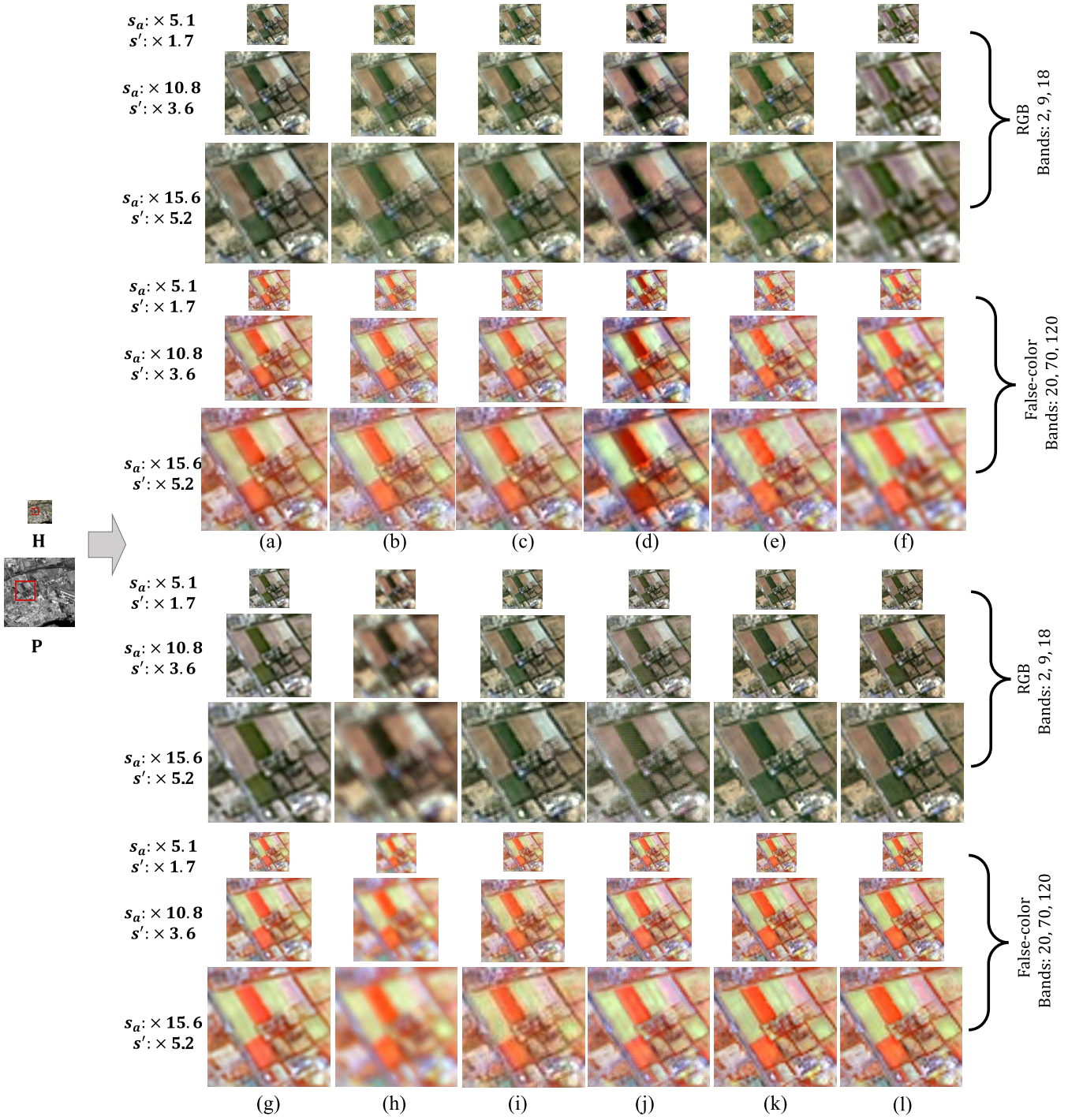


Fig. 15. Visual comparison of ARHS pansharpening on the full-resolution Changping dataset. Red boxes covering a spatial scope of 60 × 60 LRHS and a spatial scope of 180 × 180 PAN indicate the crop area for demonstration. (a) GSA. (b) SFIM. (c) MTF-GLP. (d) CNMF. (e) BayesSparse. (f) PNN-Bic. (g) HyperPNN-Bic. (h) HSpeNet-Bic. (i) ARHS-CNN. (j) VSPC-CNN. (k) NaivePINR. (l) DynamicPINR.

NaivePINR achieve overall performances relatively superior to all the other compared methods, especially for SAM, that is, our proposed methods can recover not only fine spatial details but also fine spectrum characteristics in high quality. Closer inspection can find that among traditional methods BayesSparse yields better performance than the other ones, including GSA, SFIM, MTF-GLP, and CNMF, except on the Bologna dataset, while CNN-based approaches, including PNN, HyperPNN, HSpeNet, ARHS-CNN, and VSPC-CNN, approximately produce relatively better results than those traditional methods.

In addition to quantitative analysis, we display the comparison results on the Chikusei dataset in Fig. 17. Each of the figures shows the GT and all pansharpening results of the test region in and out range of the visible bands (i.e., RGB bands and false-color bands) and is locally

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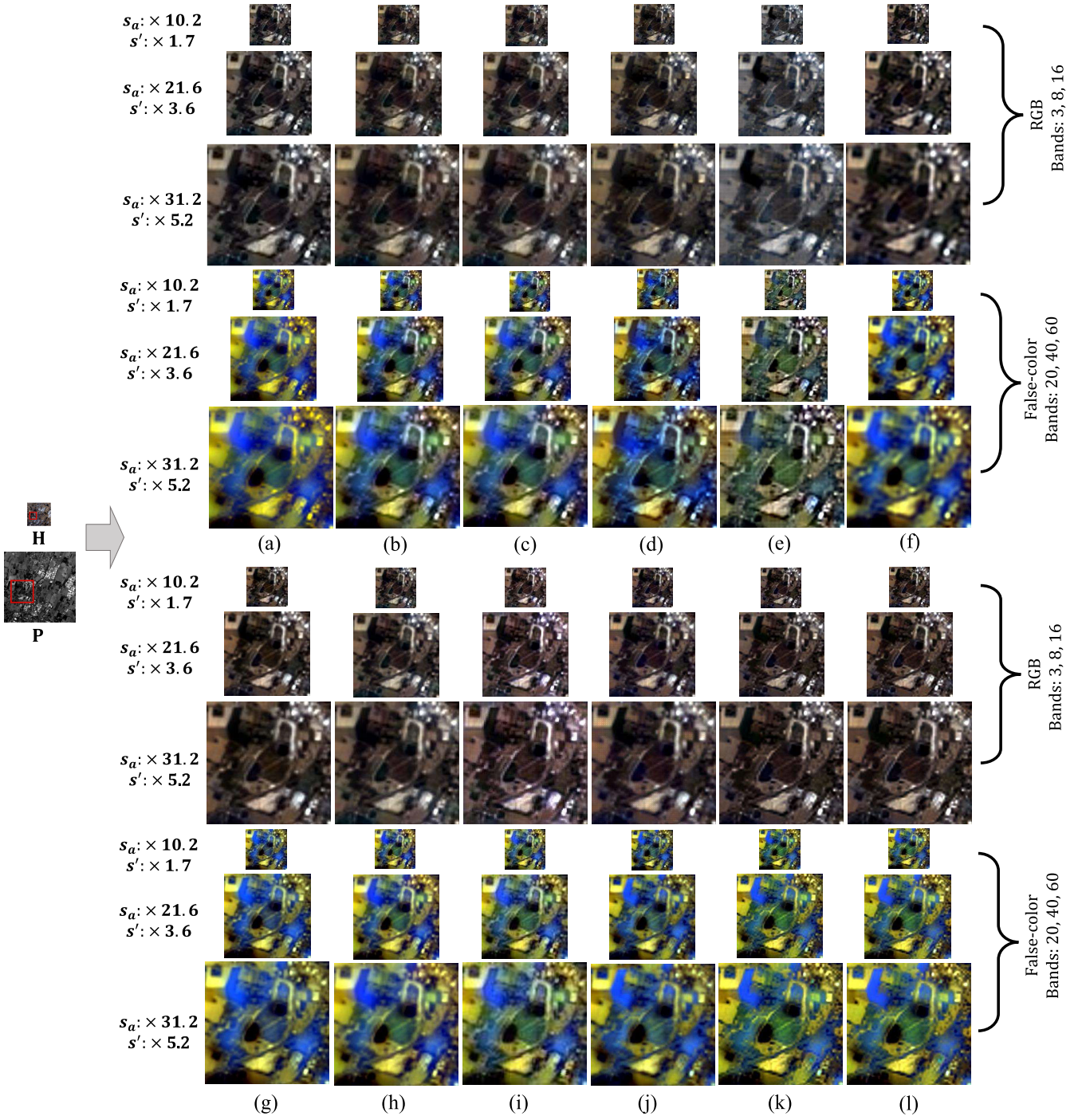


Fig. 16. Visual comparison of ARHS pansharpening on the full-resolution Bologna dataset. Red boxes covering a spatial scope of 33 × 33 LRHS and a spatial scope of 198 × 198 PAN indicate the crop area for demonstration. (a) GSA. (b) SFIM. (c) MTF-GLP. (d) CNMF. (e) BayesSparse. (f) PNN-Bic. (g) HyperPNN-Bic. (h) HSpeNet-Bic. (i) ARHS-CNN. (j) VSPC-CNN. (k) NaivePINR. (l) DynamicPINR.

amplified in red boxes with a spatial scope of 50 × 50 for better perception. As seen in the magnified areas, methods designed for standard pansharpening suffer from severe blurry effects, while ARHS pansharpening approaches successfully reconstruct fine spatial edges and details, especially for DynamicPINR and NaivePINR. Closer inspection can find that DynamicPINR recovers the more realistic color than NaivePINR, which means that the twofold dynamic tuning mechanism really works to alleviate spectral distortion. In brief, our DynamicPINR and NaivePINR tend to achieve excellent reduced-resolution results for standard HS pansharpening.

2) Full-Resolution Assessment: Full-resolution assessment, including quantitative and qualitative experiments, is conducted here to verify our proposed methods' generalization ability of standard pansharpening on real observed HS and PAN data. The quantitative experiments are performed on the two real datasets where PANs are available, i.e., Bologna

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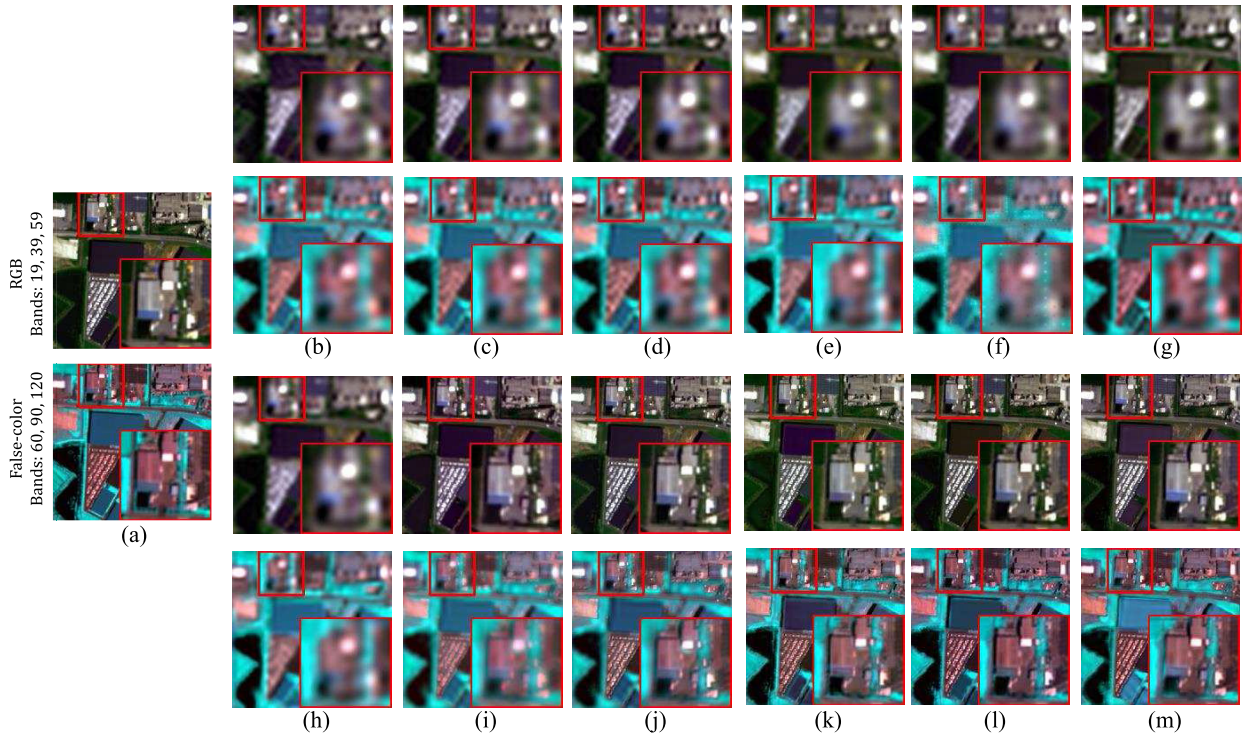


Fig. 17. Visualization of standard HS pansharpening on the reduced-resolution Chikusei dataset. Red boxes covering a spatial scope of 50×50 HS image indicate the locally amplified region for better perception. (a) GT. (b) GSA. (c) SFIM. (d) MTF-GLP. (e) CNMF. (f) BayesSparse. (g) PNN. (h) HyperPNN. (i) HSpeNet. (j) ARHS-CNN. (k) VSPC-CNN. (l) NaivePINR. (m) DynamicPINR.

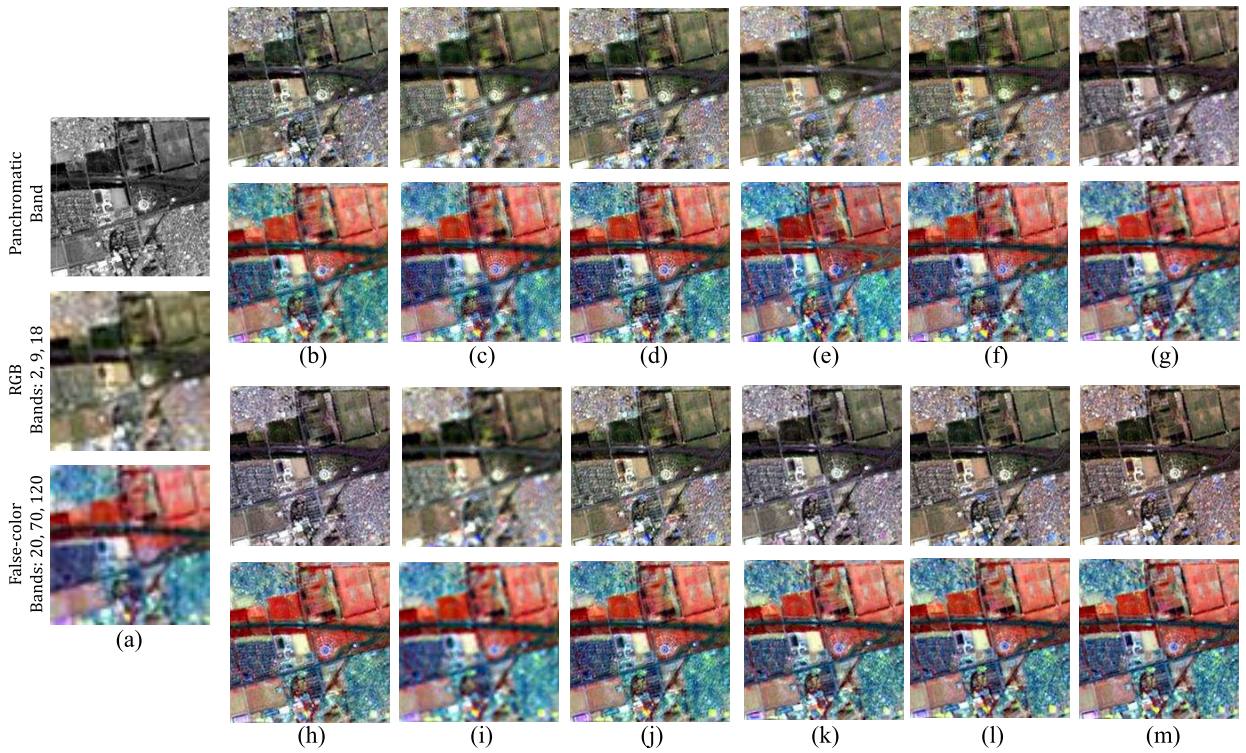


Fig. 18. Visualization of standard HS pansharpening on the real Changping dataset. (a) Observed data. (b) GSA. (c) SFIM. (d) MTF-GLP. (e) CNMF. (f) BayesSparse. (g) PNN. (h) HyperPNN. (i) HSpeNet. (j) ARHS-CNN. (k) VSPC-CNN. (l) NaivePINR. (m) DynamicPINR.

and Changping. The no-reference metrics D_λ, D_s, and QNR are then used for evaluation. Tables VI and VII numerically list the experimental results. In Tables VI and VII, we can observe that our DynamicPINR tends to gain an advantage over all the other methods, including traditional GSA, SFIM, MTF-GLP, CNMF and BayesSparse, CNN-based PNN,

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TABLE II
QUALITY METRICS OF STANDARD HS PANSHARPENING ON THE BOLOGNA DATASET

Methods	CC(↑)	SAM(↓)	RMSE(↓)	PSNR(↑)
GSA	0.8473	5.2602	0.0261	31.6526
SFIM	0.8354	5.3915	0.0275	31.2013
MTF-GLP	0.8353	5.3815	0.0278	31.0997
CNMF	0.7060	7.0983	0.0379	28.4117
BayesSparse	0.6890	18.6952	0.0984	20.1341
PNN	0.8708	4.3166	0.0217	33.2402
HyperPNN	0.8762	4.5848	0.0231	32.7293
HSpeNet	0.8795	4.2888	0.0232	32.6689
ARHS-CNN	0.8875	4.3535	0.0228	32.8242
VSPC-CNN	0.8951	4.1976	0.0212	33.4568
NaivePINR	0.8953	4.1941	0.0215	33.3253
DynamicPINR	**0.9055**	**4.1412**	**0.0209**	**33.5761**

TABLE III
QUALITY METRICS OF STANDARD HS PANSHARPENING ON THE CHANGPING DATASET

Methods	CC(↑)	SAM(↓)	RMSE(↓)	PSNR(↑)
GSA	0.8246	5.2879	0.0146	36.7101
SFIM	0.8571	4.2406	0.0124	38.0885
MTF-GLP	0.8378	4.2059	0.0135	37.3902
CNMF	0.7114	5.6107	0.0168	35.4506
BayesSparse	0.8785	4.2051	0.0117	38.5873
PNN	0.9341	4.6488	0.0085	41.3352
HyperPNN	0.9303	4.5093	0.0088	41.0445
HSpeNet	0.9156	4.2549	0.0098	40.1751
ARHS-CNN	0.9432	4.1157	0.0079	41.9804
VSPC-CNN	0.9468	4.1515	0.0076	42.2788
NaivePINR	0.9514	3.8335	0.0074	42.2865
DynamicPINR	**0.9522**	**3.7873**	**0.0073**	**42.7302**

TABLE IV
QUALITY METRICS OF STANDARD HS PANSHARPENING ON THE CHIKUSEI DATASET

Methods	CC(↑)	SAM(↓)	RMSE(↓)	PSNR(↑)
GSA	0.8902	6.3746	0.0168	35.4611
SFIM	0.8955	5.5157	0.0161	35.9068
MTF-GLP	0.8932	6.0723	0.0171	35.3247
CNMF	0.8598	6.6657	0.0208	33.6196
BayesSparse	0.9114	4.9861	0.0151	36.8031
PNN	0.8982	6.5081	0.0183	34.7292
HyperPNN	0.6072	9.4237	0.0275	31.1961
HSpeNet	0.9444	4.9011	0.0165	35.6397
ARHS-CNN	0.9518	4.7253	0.0149	36.5102
VSPC-CNN	0.9487	5.4281	0.0145	36.7152
NaivePINR	0.9548	3.9146	0.0138	37.3513
DynamicPINR	**0.9577**	**3.8093**	**0.0135**	**37.5439**

TABLE V
QUALITY METRICS OF STANDARD HS PANSHARPENING ON THE PAVIA CENTER DATASET

Methods	CC(↑)	SAM(↓)	RMSE(↓)	PSNR(↑)
GSA	0.9361	6.9066	0.0283	30.9521
SFIM	0.8928	7.9773	0.0364	28.7752
MTF-GLP	0.9535	6.1734	0.0244	32.2393
CNMF	0.9236	7.1395	0.0329	29.6491
BayesSparse	0.9627	5.8152	0.0214	33.3702
PNN	0.9575	5.8798	0.0225	32.9187
HyperPNN	0.9461	6.8578	0.0251	32.0333
HSpeNet	0.9613	5.3862	0.0222	33.0494
ARHS-CNN	0.9647	5.6459	0.0199	33.7464
VSPC-CNN	0.9703	4.6757	0.0168	35.4817
NaivePINR	0.9703	4.5919	0.0274	34.2355
DynamicPINR	**0.9798**	**4.5395**	**0.0167**	**35.8181**

TABLE VI
QUALITY METRICS OF FULL-RESOLUTION EXPERIMENT ON THE BOLOGNA DATASET

Methods	D_λ(↓)	D_s(↓)	QNR(↑)
GSA	0.0951	0.1147	0.8011
SFIM	0.0635	0.0856	0.8561
MTF-GLP	0.0862	0.1086	0.8144
CNMF	0.0747	0.1305	0.8045
BayesSparse	0.1746	0.3002	0.5775
PNN	0.0581	0.1006	0.8471
HyperPNN	0.0415	0.0777	0.8839
HSpeNet	0.0464	0.1014	0.8567
ARHS-CNN	0.0591	0.1257	0.8825
VSPC-CNN	0.0422	0.0733	0.8875
NaivePINR	0.0484	0.0588	0.8886
DynamicPINR	**0.0377**	**0.0581**	**0.9062**

TABLE VII
QUALITY METRICS OF FULL-RESOLUTION EXPERIMENT ON THE CHANGPING DATASET

Methods	D_λ(↓)	D_s(↓)	QNR(↑)
GSA	0.0716	0.1031	0.8325
SFIM	0.0372	0.0996	0.8668
MTF-GLP	0.0831	0.1639	0.7665
CNMF	**0.0256**	0.2667	0.7144
BayesSparse	0.0374	0.0794	0.8861
PNN	0.0573	0.0491	0.8963
HyperPNN	0.0661	0.0433	0.8934
HSpeNet	0.0715	0.0821	0.8898
ARHS-CNN	0.0495	0.0356	0.9068
VSPC-CNN	0.0574	0.0235	0.9105
NaivePINR	0.0484	0.0285	0.9165
DynamicPINR	0.0347	**0.0234**	**0.9243**

HyperPNN, HSpeNet, VSPC-CNN, and ARHS-CNpN, while NaivePINR achieves performances close to DynamicPINR. Besides, we performed the qualitative experiments on the real Changping dataset with real PAN data and displayed pansharpening results in Fig. 18, where pansharpened HS data are shown in and out range of visible bands (i.e., RGB bands and false-color bands). As can be seen, standard pansharpening methods suffer from severe artifact or blurry details, while four ARHS pansharpening algorithms recover obviously more clear high-frequency components contained in the observed PAN image. Closer inspection can further find that our two pansharpening INRs effectively alleviate spectral distortion, especially for DynamicPINR. In short, our DynamicPINR and NaivePINR are able to achieve satisfied quantitative and qualitative results for standard pansharpening in full-resolution assessment.

In the above ARHS pansharpening experiments and standard HS pansharpening results, we can observe the phenomenon that our DynamicPINR and NaivePINR tend to yield superior pansharpening results, which implies that our methods are able to achieve excellent pansharpening performances.

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VII. Conclusion and Future Lines

One of the main challenging problems that ARHS pansharpening is faced with is to generalize a single pansharpening model trained under certain training scales to any desired resolutions for the future HS and PAN data. Due to INRs' potential of continuous functional approximation, they offer possibilities to naturally resolve arbitrary scale generalization across training and inferring. Considering the characteristics of data involved in ARHS pansharpening, in this work, we propose two ARHS pansharpening INR methods: one is NaivePINR and the other is DynamicPINR. NaivePINR is trunk-only structured, mainly comprising a subnetwork of spectral–spatial encoding to refine spectral–spatial priors of observed scenes; and meanwhile a subnetwork of spectral–spatial query mapping to build coordinate-data connections for not only spatial coordinates but also spectral coordinates, thus reconstructing fine spectral–spatial details to fulfill ARHS reconstruction. DynamicPINR is designed to further strengthen NaivePINR. In this pansharpening INR, we develop an innovative twofold dynamic tuning mechanism to enhance the representational fidelity to specific data to refine more inherent information.

Our newly designed spectral–spatial pansharpening INRs tend to achieve excellent performances for the existing task of spatially ARHS pansharpening. Taking into consideration that our spectral–spatial INRs are built on the encoding and the query mapping not only along spatial dimension but also along a spectral dimension, they have the potentials to be conveniently extended to spectral–spatial ARHS pansharpening for some possible practical applications, that is, our spectral–spatial INRs have potentials to fuse HS images and the connected PAN images to achieve HS images with both any spatial and any spectral resolutions. In addition, some sophisticated factors, such as various degradations and noise effects, were not explicitly considered in our work. However, they indeed deserve considerations for the practical applications of HS pansharpening technique. Therefore, designing spectral–spatial pansharpening INRs, which comprise specific structures to handle noise effects and various degradations, could also be a future research direction.

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Lin He (Member, IEEE) received the B.S. degree from Xi'an Institute of Technology, Xi'an, China, in 1995, the M.S. degree from Chongqing University, Chongqing, China, in 2003, and the Ph.D. degree in pattern recognition and intelligent systems from Northwestern Polytechnical University, Xi'an, in 2007.

Since 2007, he has been with the School of Automation Science and Engineering, South China University of Technology, Guangzhou, China, where he is currently a Full Professor. His research interests include unmanned aerial vehicle (UAV) image processing, hyperspectral image processing, spectral–spatial–temporal information fusion, machine learning, and high-dimensional signal processing.

Dr. He is serving as an Associate Editor for IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and IEEE JOURNAL ON MINIATURIZATION FOR AIR AND SPACE SYSTEMS.



Zhou Fang received the B.S. degree from the School of Information Science and Engineering, China University of Petroleum, Beijing, China, in 2022. He is currently pursuing the M.S. degree with the School of Automation Science and Engineering, South China University of Technology, Guangzhou, China.

His research interests include degraded information reconstruction from hyperspectral images, data fusion, and machine learning in remote sensing.

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Jun Li (Fellow, IEEE) received the B.S. degree in geographic information systems from Hunan Normal University, Changsha, China, in 2004, the M.E. degree in remote sensing from Peking University, Beijing, China, in 2007, and the Ph.D. degree in electrical engineering from the Instituto de Telecomunicações, Instituto Superior Técnico (IST), Universidade Técnica de Lisboa, Lisbon, Portugal, in 2011.

From 2013 to 2021, she was a Full Professor with Sun Yat-sen University, Guangzhou, China. Since 2022, she has been with China University of Geosciences, Wuhan, China, as a Full Professor. She has obtained several prestigious funding grants at the national and international levels. She has published more than 160 journal citation report (JCR) articles, 60 international conference papers, and a book chapter.

Dr. Li has been serving as the Editor-in-Chief for IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING since 2021.



Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

Since 1999, he has been with Grenoble INP, where he is currently a Professor of signal and image processing. He has been a Visiting Scholar at Stanford University, Stanford, CA, USA; the KTH Royal Institute of Technology, Stockholm, Sweden; and the National University of Singapore (NUS), Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavík, Iceland. He holds the AXA Chair in remote sensing and is an Adjunct Professor at the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008 and the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He is a member of the Institut Universitaire de France from 2012 to 2017. He was the Founding President of the IEEE Geoscience and Remote Sensing French Chapter from 2007 to 2010, which received the 2010 IEEE GRS-S Chapter Excellence Award. He has received multiple outstanding paper awards. He was the Vice-President of the IEEE Geoscience and Remote Sensing Society, in charge of meetings and symposia, from 2017 to 2019. He was the General Chair of the first IEEE GRSS Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote Sensing (WHISPERS). From 2005 to 2008, he was the Co-Chair of the GRS Data Fusion Technical Committee, where he was the Chair from 2009 to 2011. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON IMAGE PROCESSING, and PROCEEDINGS OF THE IEEE. He was the Editor-in-Chief of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. In 2014, he served as a Guest Editor for *IEEE Signal Processing Magazine*. He has been a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters) since 2018.



Antonio Plaza (Fellow, IEEE) received the M.Sc. and Ph.D. degrees in computer engineering from the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura, Cáaceres, Spain, in 1999 and 2002, respectively.

He is currently a Full Professor and the Head of the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura, Cáaceres, Spain. He has authored more than 600 publications, including 284 JCR journal articles (more than 170 in IEEE journals), 24 book chapters, and over 300 peer-reviewed conference proceedings papers. He has guest edited ten special issues on hyperspectral remote sensing for different journals. His main research interests comprise hyperspectral data processing and parallel computing of remote sensing data.

Dr. Plaza was a member of the Editorial Board of the IEEE Geoscience and Remote Sensing Newsletter from 2011 to 2012 and *IEEE Geoscience and Remote Sensing Magazine* in 2013. He was also a member of the Steering Committee of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS). He is a fellow of IEEE for his contributions to hyperspectral data processing and parallel computing of Earth observation data. He was a recipient of the recognition of Best Reviewers of IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2009 and the recognition of Best Reviewers of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2010, for which he served as an Associate Editor from 2007 to 2012. He was a recipient of the Best Column Award of the IEEE Signal Processing Magazine in 2015, the 2013 Best Paper Award of the JSTARS journal, and the most highly cited paper for the term of 2005–2010 in *Journal of Parallel and Distributed Computing*. He received the Best Paper Awards at IEEE International Conference on Space Technology and the IEEE Symposium on Signal Processing and Information Technology. He served as the Director of Education Activities for the IEEE Geoscience and Remote Sensing Society (GRSS) from 2011 to 2012 and the President for the Spanish Chapter of IEEE GRSS from 2012 to 2016. He has reviewed more than 500 manuscripts for over 50 different journals. He served as the Editor-in-Chief for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING from 2013 to 2017. He is also an Associate Editor of IEEE ACCESS. He is included in the 2018 and 2019 Highly Cited Researchers List (Clarivate Analytics). Additional information: http://www.umbc.edu/rssipl/people/aplaza.

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